

Acknowledgment

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Glossary of terms and acronyms

ESP Electric submerged pump

API American Petroleum Institute

HP High pressure

HP Horse Power

IPR : Inflow Performance Relationship

TPR :Tubing Performance Relationship

rcf : cubic feet of reservoir volume

scf : standard cubic foot

md: millidarcy

cP: centipoise

RPM: rotation per minute

BEP: best efficiency point

UNITS

API American Petroleum
 Institute crude grade
 $API = (141.5 / \text{Specific gravity}) - 131.5$
 $\text{Spec gravity} = 141.5 / (API + 131.5) \text{ kg/l}$
 Bl Barrel (of oil) 1 Bl = 42 Gallons
 1 Bl = 159 liters
 1 Bl equiv. to 5487 scf = 147 scm gas
 Bpd Barrel per day 1 Bpd $\approx$ 50 tons/tons per year
 BTU British thermal unit 1 BTU = 0.293 Wh = 1,055 kJ
 Cal Calorie 1 Cal = 4,187 J (Joules)
 MMscf Million standard cubic
 feet
 1 MMscf = 23.8 TOE $\approx$ 174 barrels
 psi Pounds per square
 inch
 1 psi = 6.9 kPa = 0.069 atm
 Scf Standard cubic feet
 (of gas) defined by
 energy, not a normalized
 volume
 1 scf = 1000 BTU = 252 kcal
 = 293 Wh = 1,055 MJ
 $\approx$ 0.0268 scm
 Scm Standard cubic meter
 (of gas, also Ncm)
 Defined by energy
 content
 1 Scm = 39 MJ = 10.8 kWh
 1 Scm $\approx$ 37.33 Scf (not a volume conv.)
 1 Scm $\approx$ 1.122 kg
 TOE Tons oil equivalent
 Range 6.6 - 8 barrels at
 API range 8 - 52
 1 TOE = 1000 kg = 1 Ton (metric) oil
 1 TOE = 1 Tone oil (US)
 1 TOE $\approx$ 7.33 Barrels (at 33 API)
 1 TOE $\approx$ 42.9 GJ = 11.9 MWh
 1 TOE $\approx$ 40.6 MMBTU
 1 TOE $\approx$ 1.51 ton of coal
 1 TOE $\approx$ 0.79 ton LNG
 1 TOE $\approx$ 1,125 Scm = 42,000 Scf
 kWh Kilowatt hour
 = 1000 joules * 3600 S
 1 kWh = 3.6 MJ = 860 kcal = 3,413 BTU

General Introduction

This report entitled "ESP Design Using Hand Sizing and AUTOGRAPHPC for x wells" incorporates the graduation project in the National Engineering School of Monastir elaborated in the international oil and gas company OMV.

Oil and gas sector is defined by **OPEC**(organization of petroleum countries) as the engine of the world economy. It is an essential source of energy in numerous domains such as (transportation, industries, home and medicine ...); it has shaped our world in many important ways.

The amount of oil and gas extraction is dependent on well productivity. Unfortunately only 5% of worldwide wells flow naturally which involves the application of artificial techniques of well activation.

The vitality of oil and gas industry generates inventions in terms of artificial lifting in order to optimize the production and maximize the oil companies profitability.

The electrical submersible pump renowned for ESP is an efficient form of artificial lift utilized for lifting moderate to high amounts of fluids (hydrocarbons) with high water cut.

The main objective of this work is to design the ESP for **X wells (Wi....) of X** field through 2 methods: the nine hand sizing steps and the **software** method to come up with meaningful sensitivity study in order to expect the ESP contribution for the forthcoming installation.

The first chapter is a shortly presentation of the OMV company, then chapter two and three contain a literature review to be familiar with basic concept of production engineering such as performance of petroleum system and artificial technique of activation with braking the lights on ESP features. The forth chapter is the real case study of ESP design on **X well(s)** of **X** field, the fifth chapter leads to **X** software used to computerized design of the ESP to sum up the results of sensitivity analysis. Finally the economic study is done in chapter six to know the profitability of the ESP.

Chapter1: Company presentation

1. General description

OMV("österreichische mineralölverwaltung") is an integrated international oil and gas company, headquartered in Vienna. OMV's main business is in Exploration & Production (E&P), Gas & Power (G&P) and Refining & Marketing (R&M). With group sales of more than **X billion** and a global workforce of around X in 2016, OMV is the largest listed manufacturing company in Austria. In 2016, OMV produced around **X tonnes** of oil and **X billion** cubic meters of gas in Austria.

2. Objective

The main objectives of OMV are:

- Manage exploration and production of hydrocarbon on behalf of the Tunisian government.
- Producing oil that will allow Tunisia to accelerate its economic development and establish position on the world stage.

3. Field of activity

- ✓ Exploration and production of hydrocarbon
- ✓ Marketing of crude oil and petroleum products
- ✓ oil Service
- ✓ Human Resources Development

Chapter2: Oil and Gas general overview

1. Introduction

Petroleum production engineering is the series of activities concerned with the ability of a well to produce. These activities are divided, according their specification, into upstream sector and downstream sector:

- The upstream sector, it mainly consists in searching and finding oil from underground(underwater) fields called "Exploration" followed by oil or gas extraction "Production".

Exploration

Reservoir capacity is determined approximately by two different methods:

✓ **Geological survey:** In this case geoscientists attempt to locate best areas by examining different underground layers of rock using advanced technologies and special tools such as "aerial photography, satellite pictures and specialized machines that measure variation in the earth's gravity fields."

✓ **Seism survey** are a vital part in oil and gas exploration, it involve using sound waves to form a clear picture of underwater layer rock. Surveyors register the seismic waves that are produced by an explosion or sound generator. This technique is based on determinations of the time interval that elapses between the initiation of a seismic wave at a selected shop point and the arrival of reflected or refracted impulses at one or more seismic detectors.

After designating the specific area based on geoscientists investigation the second step should be begin which is the initial drilling. If drilled wells, called wildcat well, **present good signs** of existing crude oil inside then the well will be completed and the second phase of production starts.

Production

Once oil is found after the preliminary exploration phase and the drilling of exploratory wells, the production phase can begin: wells are perforated adopting the rotary rig drilling's technique. During this operation underground water should be protected from oil and gas contamination, thus outer casings must be inserted in the well then cemented to the exterior wall. Once the reservoir rock is reached the well is completed with the remaining facilities: wellhead and surface equipments.

- The downstream sector: The downstream sector is the part of the oil industry involved with purifying crude oil and refining it into different products. It also involves the transportation and marketing of crude oil and its products.

Once oil starts flowing, it needs to be extracted in large volumes and then taken to special sites where it is treated carefully before being transported internationally. The process through which crude oil is purified and treated to remove unusable substances is called **REFINING**. This process is also used to separate oil into different usable petroleum products. All this takes place in an oil refinery.

All refineries perform three basic functions: Separation Conversion Treatment.

Refineries are highly automated and technologically advanced. That is why a typical refinery costs billions of dollars to build. It also costs millions to operate, employing hundreds of people and running every day of the year.

SEPARATI ON

In this phase, a refinery heats crude oil to different temperature levels. Different parts of crude oil have different boiling points. As the temperature rises, these different parts or fractions are separated. This is done inside distillation towers.

Conversion

In this phase, high temperatures and pressure, as well as chemical catalysts, are used to 'crack' or split heavy hydrocarbon molecules into smaller, more desirable ones. This is the most widely used conversion method and it is called CRACKING.

Treatment

The final phase is treatment. In this step, the fractions produced during separation are treated to improve their quality. They are then blended with other elements to produce the final products.

2. Presentation of the petroleum production system:

2.1. Definition

Petroleum production involves two distinct but intimately connected general systems: the reservoir, which is a porous medium with unique storage and flow characteristics; and the artificial structures, which include the well, bottom hole, and wellhead assemblies, as well as the surface gathering, separation, and storage facilities.

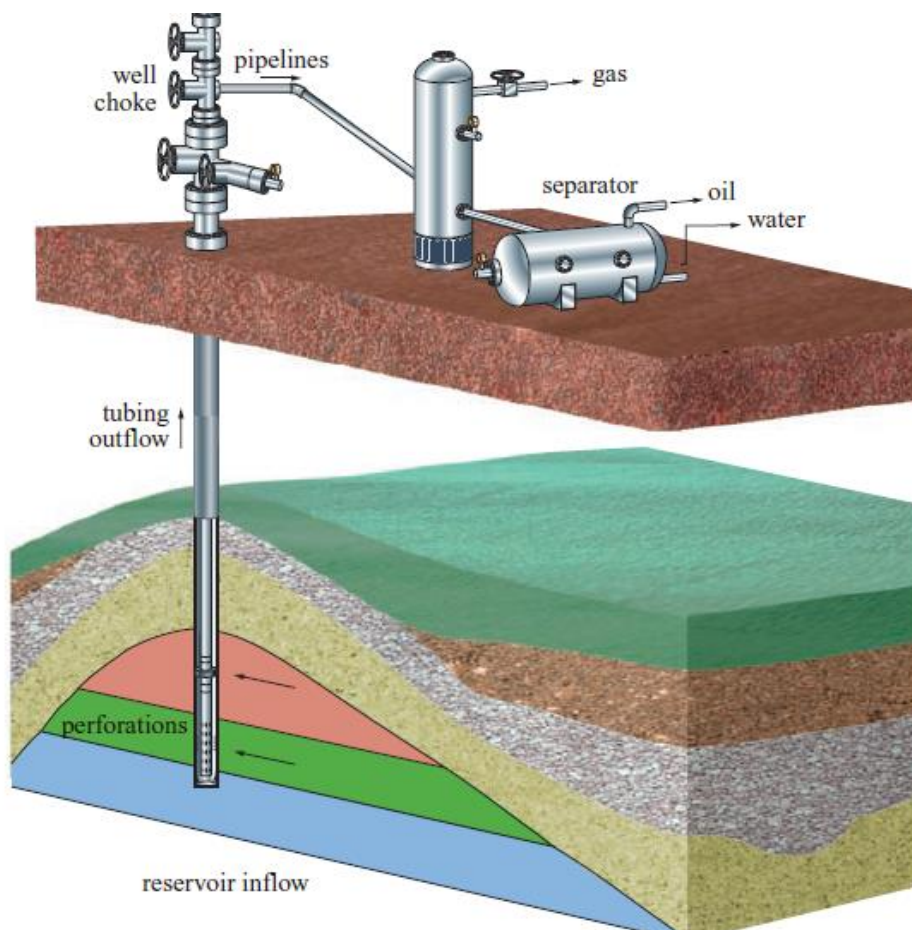


Figure 1 Petroleum Production System

2.2. Pressure losses across the petroleum production system

The hydrocarbon fluid streams from the reservoir into the well, up the tubing, along the horizontal flow line and into the oil storage tank. During this process the fluid's pressure is reduced from the reservoir pressure to atmosphere pressure in a series of pressure loss processes (Figure 2):

These pressure losses can be classified into three main components:

- the total pressure losses in the reservoir and completion
- the total pressure losses in the tubing
- the total pressure losses at the surface

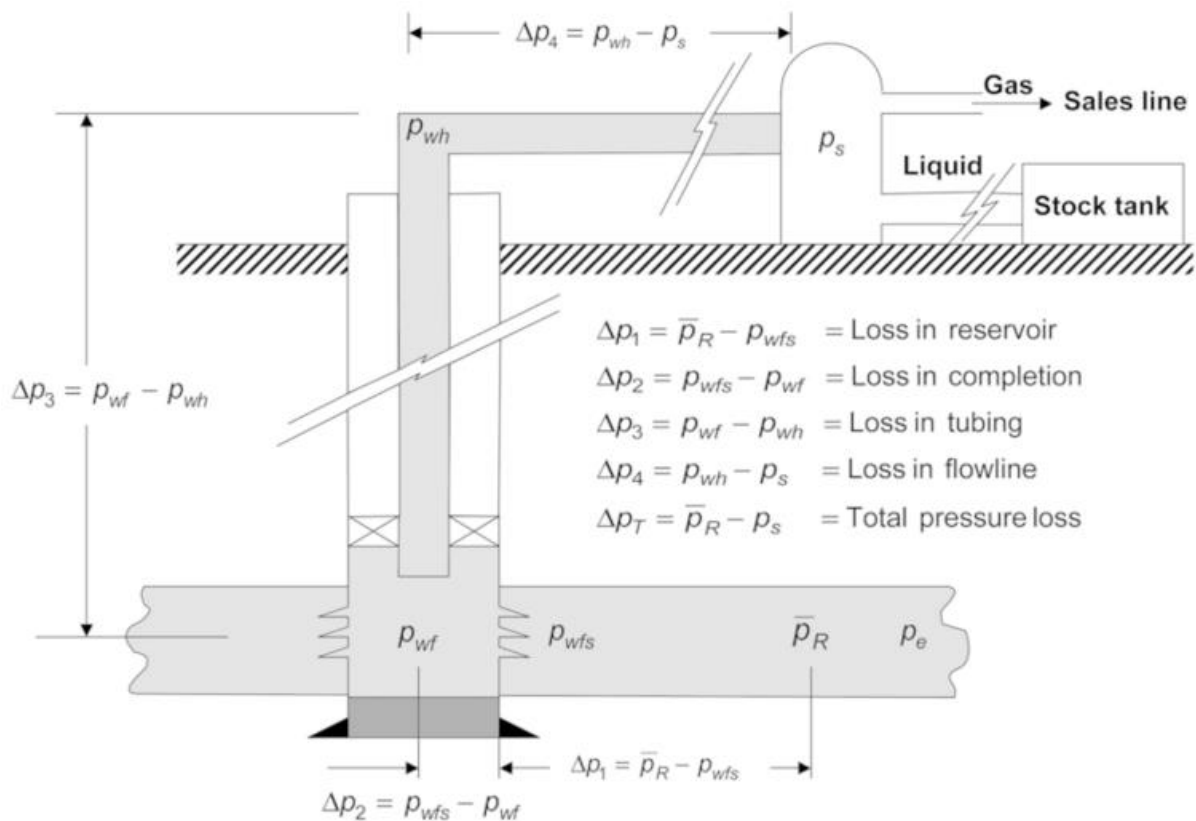


Figure 2: main pressure losses within production system

3. Petroleum production engineering

3.1. Introduction

Production engineering technologies attempt to maximize oil and gas production in a most profitable way possible. It offers different methods and technologies allowing to:

- ✓ Evaluate inflow and outflow performance between the reservoir and the wellbore.
- ✓ design completion system
- ✓ Select the proper artificial lift equipment
- ✓ Select equipment for surface facilities

The only way to achieving these previous responsibilities, is for production engineers to elaborate a detailed analysis of these distinct, yet related parts:

- The components of oil and gas production system
- The fundamentals of well performance
- well completion
- oil wells activation systems

3.2. Production's technologies: optimization and performance

3.2.1. Introduction

Well analysis is the most important step to optimize oil production. The goals of production optimization are to find the flow rate at which well will produce with given completion configurations and conclude if it needs exterior interventions.

various approaches have been proposed, over the years on required operating conditions to prepare the well for an optimal production rate. Multivariable optimization tools and techniques have been utilized in many aspects of the oil industry to support decisions in accomplishing tasks in production engineering, such as designing production systems and facilities.

The derivability of a well can often be influenced by the performance of one component in the system, which makes it really important to analyze the well to allow the determination of the producing capacity for any combination of components. This requires a system analysis approach often called 'NODAL' analysis. It involves employing correlations to predict multiphase flow behavior through pipes, well completions, restrictions and the reservoir.

To apply the system analysis procedure to a well, it is necessary to be able to calculate the pressure drop that will occur in all the system's components.

Well inflow performance relationship (IPR) and tubing performance relationship (TPR) are the basic requirements for well analysis. Accurate well tests and flowing data must be obtained and then the proper IPR and two phase flow correlation models could be applied for successful analysis. Then mathematical models of other well components can be used to complete the predicted well performance.

- The inflow performance:

The flow from the reservoir into the well has been called "inflow performance" by Gilbert and a plot of producing rate versus bottom-hole flowing pressure is called an "inflow performance relationship" or IPR.

- The tubing performance relationship:

The pressure loss in tubing is a difficult value to predict because it's a function of many factors that are not always available; other than the tubing size and flow rate, there are fluid densities, fluid viscosities, temperature gradient, gas liquid ratio, water oil ratio and slippage.

3.2.2. Well productivity

An oil or gas reservoir contains highly compressible hydrocarbon fluids at an elevated

pressure and temperature and as such, the fluid stores up within itself considerable energy of compression. The efficient production of fluids from a reservoir requires the effective dissipation of this energy through the production system. Optimum utilisation of this energy is an essential part of a successful completion design and ultimately of field development economics. Where necessary and economic, this lift process can be supported by artificial lift using pumps or gas lift.

The productivity of the system is dependent on the pressure loss which occurs in several areas of the flow system namely:-

- The reservoir
- The wellbore
- The tubing string
- The choke
- The flow line
- The separator

These are shown in figure 4. Under natural flowing conditions the reservoir pressure must provide all the energy to operate the system i.e. all the pressure drop in the system.

$$P_R = \Delta P_{\text{SYSTEM}} + P_{\text{SEP}}$$

where;

P_R = reservoir pressure

ΔP_{SYSTEM} = total system pressure drop

P_{SEP} = separator pressure

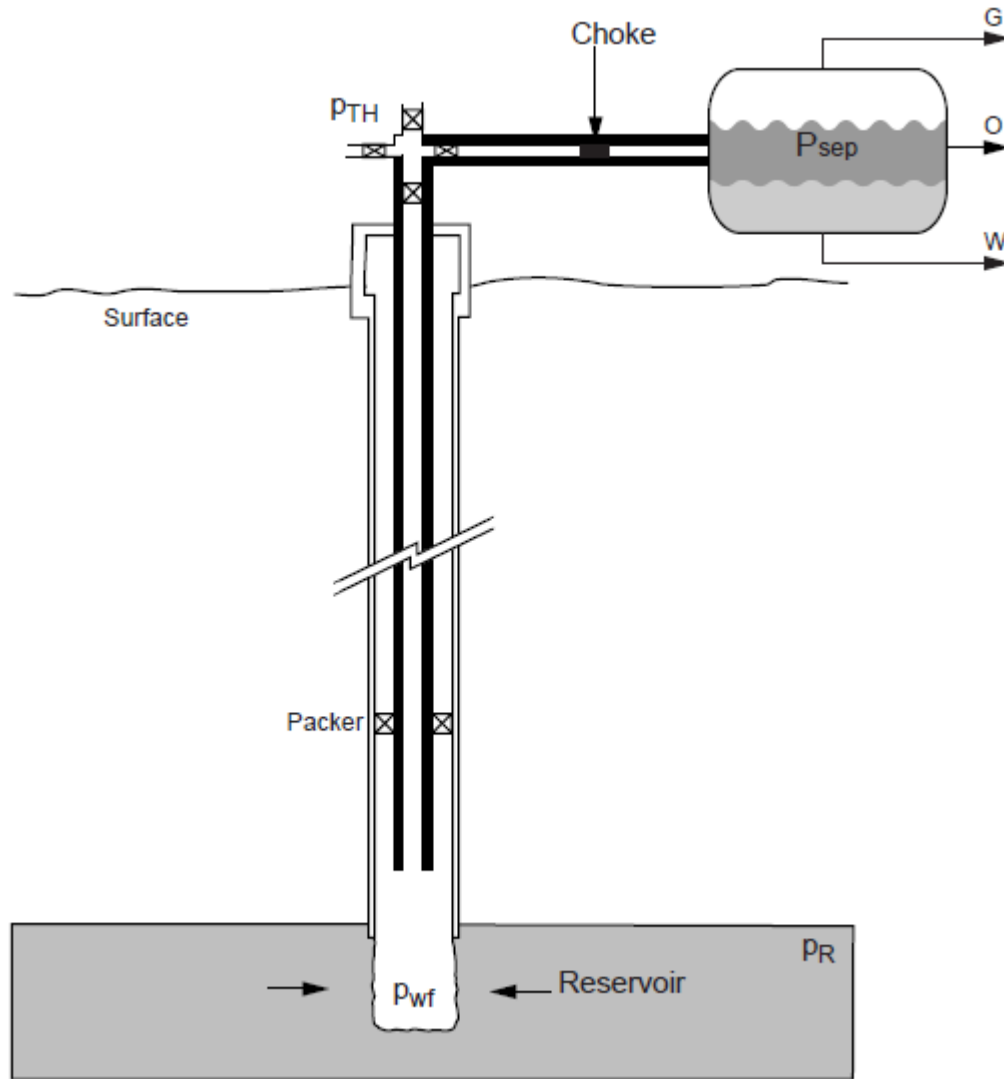


Figure 3 The production system

The optimum distribution of energy between these various areas has a major bearing on the cost effectiveness of a well design and hence production costs.

The pressure drop which occurs across the reservoir, ΔP_{RES} and is defined as the ***inflow performance relationship*** or ***IPR***. The pressure drop and causes flow is in the tubing and wellbore ΔP_{TBG} is that which occurs in lifting the fluids from the reservoir to the surface and it is known as the ***vertical lift performance*** or ***VLP***, or ***the tubing performance relationship*** or ***TPR***,

i.e. for natural flow $R = \Delta P_{RES} + \Delta P_{TBG} + P_{TH}$

Where;

P_{TH} = Tubing head pressure

The pressure drop across the reservoir, the tubing and choke are rate dependant and these relationships therefore define the means by which we can optimise the production of the fluid from the reservoir.

3.2.3. Well completion

Historically the major proportion of production technology activities have been

concerned with the engineering and installation of the down hole completion equipment.

The completion string is a critical component of the production system and to be effective it must be efficiently designed, installed and maintained. Increasingly, with moves to higher reservoir pressures and more hostile development areas, the actual capital costs of the completion string has become a significant proportion of the total well cost and thus worthy of greater technical consideration and optimisation. The completion process can be split into several key areas which require to be defined including:-

- (1) The fluids which will be used to fill the wellbore during the completion process must be identified, and this requires that the function of the fluid and the required properties be specified.
- (2) The completion must consider and specify how the fluids will enter the wellbore from the formation i.e., whether in fact the well will be open or whether a casing string will be run which will need to be subsequently perforated to allow a limited number of entry points for fluid to flow from the reservoir into the wellbore.
- (3) The design of the completion string itself must provide the required containment capability to allow fluids to flow safely to the surface with minimal loss in pressure. In addition however, it would be crucial that the string be able to perform several other functions which may be related to safety, control, monitoring, etc. In many cases the completion must provide the capacity for reservoir management. The completion string must consider what contingencies are available in the event of changing fluid production characteristics and how minor servicing operations could be conducted for example, replacement of valves etc.

3.2.3.1. Type of well completion

- Open hole completion
- open hole cavitations' process
- cased-hole completion

3.2.4. Well stimulation

The productivity of a well naturally arises from the compressed state of the fluids, their mobility and the flow properties of the rock, primarily in terms of permeability. In some cases reservoirs may contain substantial reserves of hydrocarbons but the degree of inter-connection of the pore space and the ease with which the fluids can flow through the rock, may be very poor. In such situations it may be beneficial to stimulate the production capacity of the well. Stimulation techniques are intended to:-

- (1) Improve the degree of inter-connection between the pore space, particularly for low permeability or vugular rocks
- (2) Remove or bypass impediments to flow, e.g., damage.
- (3) Provide a large conductive hydraulic channel which will allow the wellbore to communicate with a larger area of the reservoir.

In general, there are four principal techniques applied, namely:-

- (1) **Propped Hydraulic Fracturing** - whereby fluids are injected at a high rate and at a pressure which exceeds the formation break down gradient of the formation. The rock will then fail mechanically producing a "crack". To prevent closure or healing of the fracture, it is propped open by a granular material. This

technique increases the effective well bore radius of the well.

Matrix Acidisation - this process is conducted at pressures below the formation break down gradient and requires the injection of acid into the reservoir to either dissolve the rock matrix and/or dissolve damage material contaminants which has invaded the rock pore space. The main objective of acidisation is to increase the conductivity of the rock.

(3) **Acid Fracturing** - whereby acid injected at a pressure above the formation breakdown gradient, creates a fracture. The acid then etches flow channels on the surface of the fracture which on closure will provide deep conductive flow channels.

(4) **Frac Packing** - which is a shallow penetrating hydraulic fracture propagated usually into a formation of moderate to high permeability, and is subsequently propped open prior to closure. The process is used to reduce the near wellbore flow induced stress, and in some cases can also limit/reduce sand production. A number of other chemical treatments are available for specific situations.

3.2.5 Well performance (System nodal analysis)

globalspec: "A typical gas well may have to flow against many flow restrictions in order for the produced gas to reach the surface separator. The gas must flow (1) through the reservoir rock matrix, (2) then through the perforations and possible gravel pack, (3) possibly through a bottomhole standing valve, (4) through the tubing, (5) possibly a subsurface safety valve, and (6) through the surface flowline and flowline choke to the separator. Each of these components will have a flow-dependent pressure loss. A change in any of the well restrictions will affect the well production rate. To determine overall well performance, all components of the well must be considered as a unit or total system.

One useful tool for analyzing well performance is Systems Nodal Analysis. Nodal Analysis divides the total well system into two subsystems at a specific location called the "nodal point." One subsystem considers the inflow from the reservoir, through possible pressure drop components, and to the nodal point. The other subsystem considers the outflow system from some pressure on the surface down to the nodal point. For each subsystem, the pressure at the nodal point is calculated and plotted as two separate, independent pressure-rate curves.

<http://www.globalspec.com/reference/28804/203279/chapter-4-systems-nodal-analysis>

"

petrowiki "The objective of systems analysis is to combine the various components of the production system for an individual well to estimate production rates and optimize the components of the production system."

Systems analysis, which has been applied to many types of systems of interacting components, consists of selecting a point or node within the producing system (well and surface facilities). Equations for the relationship between flow rate and pressure drop are then developed for the well components both upstream of the node (inflow) and downstream (outflow). The **flow rate** and **pressure** at the **node** can be calculated since:

(i) Flow into the node equals flow out of the node.

(ii) Only one pressure can exist at the node.

Further, at any time, the pressure at the end points of the system {separator (P_{sep}) and reservoir pressure (P_R)} are both fixed. Thus:

$P_R - (\text{Pressure loss upstream components}) = P_{node} \quad (1)$

$P_{sep} + (\text{Pressure loss downstream components}) = P_{node}$

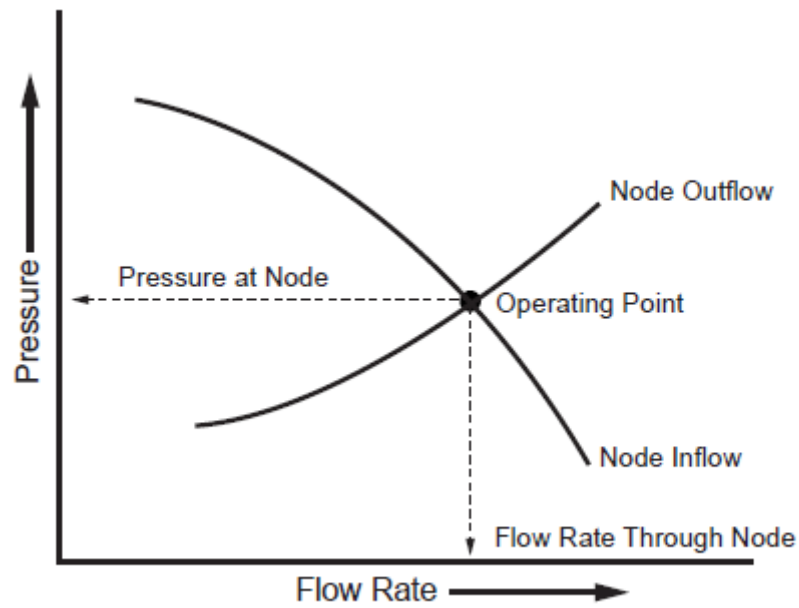


Figure 4 Node flow rate and pressure

3.2.5.1 Inflow performance from the reservoir to the wellbore

3.2.5.1.1 Introduction

The proper description of well behavior requires that the relationship between bottom-hole pressure and the corresponding production rates be established. The resulting function is called the well's inflow performance relationship (IPR) and is usually obtained by running well tests.

3.2.5.1.2 Productivity Index concept

It is the most optimistic approach to describe the inflow performance of oil wells.

To utilize this concept, four assumptions have to be realized:

- Radial flow near the wellbore area
- A single phase, incompressible liquid is flowing
- A homogeneous distribution of the formation permeability
- The fluid is fully saturated in the formation

For general flow through porous media:

$$Q = \frac{KA(P_0 - P_1)}{\mu L}$$

But in our case we're working with oil reservoirs to find the production rate of any oil well or the Darcy law equation:

$$Q = \frac{7.08 \cdot 10^{-3} h K_0 (P_r - P_{wf})}{B_0 \mu_0 \ln\left(\frac{r_e}{r_w}\right) - 0.75}$$

where

B_0 : liquid volume factor, bbl/STB

μ_0 : average viscosity, cP

r_e : drainage radius of well, ft

r_w : radius of wellbore, ft

K_0 : effective permeability, md

h : effective feet of pay(height), ft

P_r : reservoir pressure, psi

P_{wf} : flowing bottom-hole pressure, psi

If we make the assumption that h, K_0, r_w, r_e, μ_0 and B_0 are constant for a particular well the equation becomes:

$$q = K(P_R - P_{wf})$$

where K is the Productivity Index

finally we obtain the "equation2"

$$q = PI(P_R - P_{wf})$$

PI is usually found by measurement (down-hole gauge and surface flow rate).

It calculates the highest maximum flow rate(AOF) since no change from producing below bubble point is assumed.

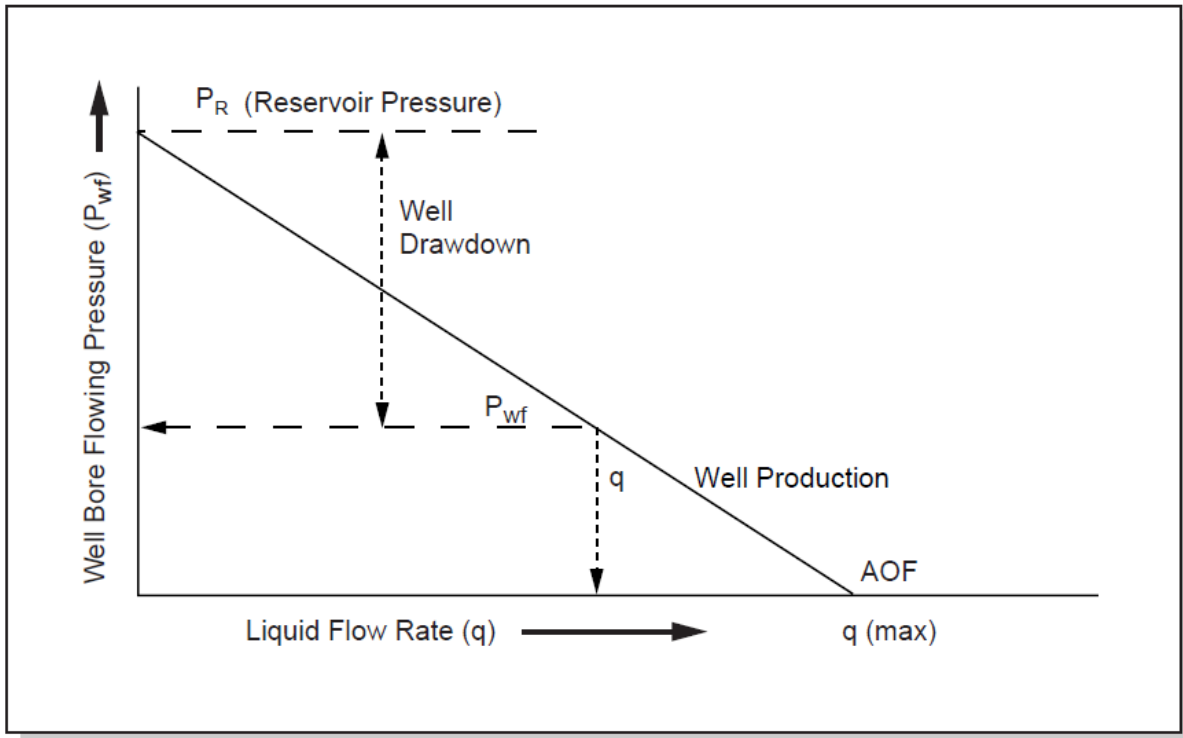


Figure 5 Straightline IPR (for an incompressible liquid)

We can notice that the curve of the wellbore flowing pressure (P_{wf}) in function of the flow rate (q) is a straight line of a negative slope ($-1/PI$). Also the graph shows : the

lowest point of the curve represents the maximum of the potential rate corresponding to the minimum of the wellbore flowing pressure which is zero whereas the highest point matching the two values of the minimum flow rate (zero) and the maximum wellbore flow rate (P_R : reservoir pressure) that can be attained.

the maximum flow rate which is impossible to achieve is called typically Absolute Open Flow Potential typically known for the abbreviation AOF. This latter is used only to compare between different wells' deliverability. So to obtain the flow rate at any flowing bottom-hole pressure it's sufficient to know the productivity index PI , the bottom-hole pressure P_{wf} and apply the "equation 2". the productivity index is defined as the flow rate per unit pressure drop.

3.2.5.1.3 Vogle's method

Straight line IPR (Section 1.4.1) are also not applicable to when two phase inflow is taking place, e.g. when saturated oil is being produced. Vogel ("Inflow Performance Relationships in Solution-Gas Drive Wells", J Pet Tech, 1968, 83-92) proposed the following equation based on a large number of well performance simulations:

$$\frac{q}{q_{\max}} = 1 - 0.2\left(\frac{P_{wf}}{P_R}\right) - 0.8\left(\frac{P_{wf}}{P_R}\right)^2$$

where q: liquid rate

$q_{\max}$ is the AOF

PR: reservoir pressure

Pwf: bottom-hole flowing pressure

"Vogel studied the inflow performance of the well with a solution gas as a drive mechanism. He evaluated several cases and scenarios where pressure is below bubble point pressure. He kept changing different parameters like fluid and rock properties. At the end he came up with a conclusion stating that all IPR curves have the same shape for these different scenarios (Figure X : Vogel's inflow performance curve [x]). This conclusion can be represented by the following dimensionless equation(Eq X)

$$\frac{q}{q_{\max}} = 1 - 0.2(P_{wf} / P_R) - 0.8(P_{wf} / P_R)^2$$

where q: production rate at bottom-hole pressure (Pwf), STB/d

Qmax: maximum production rate, STB/d

Pr: average reservoir pressure, psi

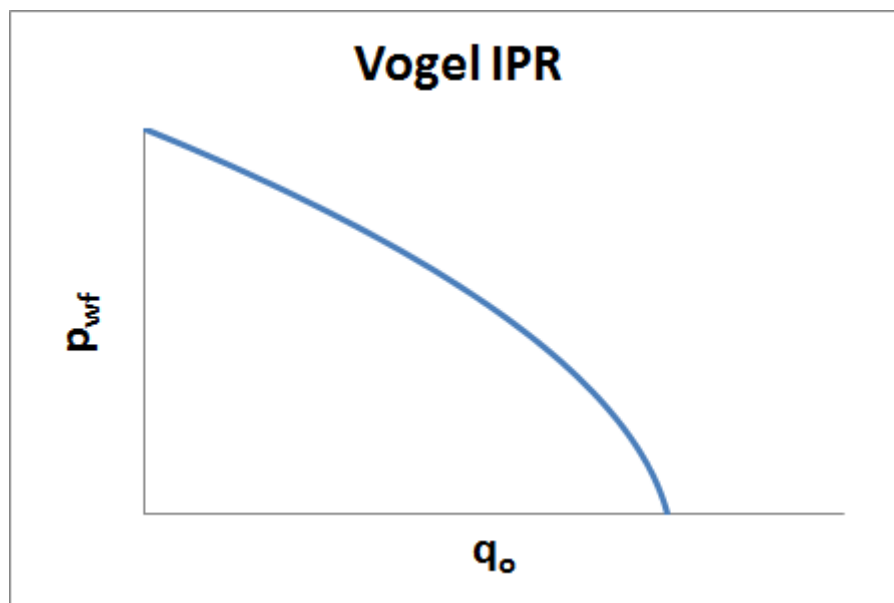


Figure 6: Vogel 's inflow performance curve

Eq x is applied when reservoir pressure is below bubble point and oil is the only fluid produced.

3.2.5.1.4 Sum up IPR

When multirate test data is available then equation [7] is preferred since it includes high rate (non-Darcy or turbulent) effects. This is best done by plotting the data in a similar manner to Figure 8, the resulting straight line has a slope of $1/n$.

Figure 9 compares the production rate as a function of drawdown for an under-saturated

oil (straight line IPR, line A) and a saturated oil showing the two phase flow effects discussed above (curve B). The figure also shows the special case (curve C) when the wellbore pressure is below the bubble point while the reservoir pressure is above, i.e. (incompressible) liquid flow is occurring in the bulk of the reservoir.

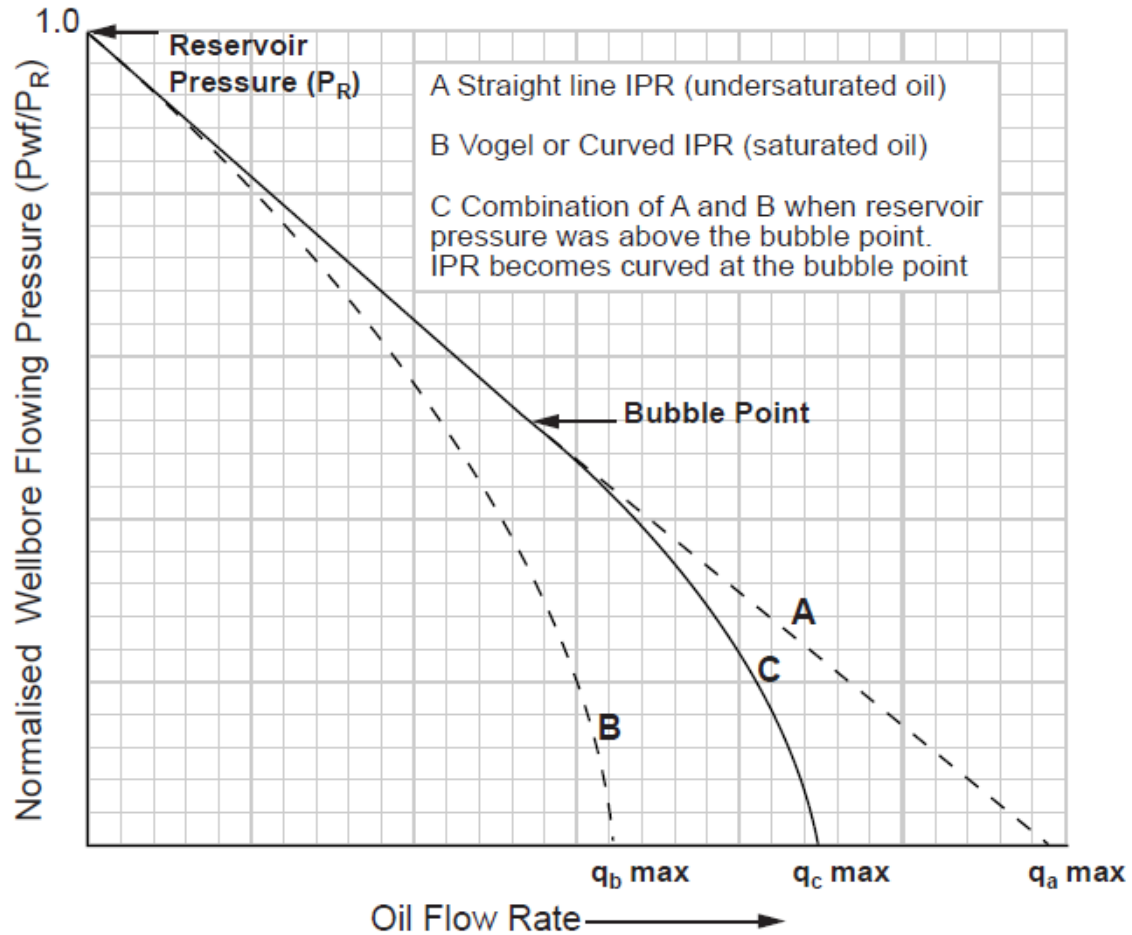


Figure 7: Inflow Performance Relationship

1.2.5.1 Outflow performance of oil and gas well from the wellbore to the surface.

1.2.5.1.1 Introduction

Just as there is a drop in pressure within the formation during production, there is also a drop in pressure within the tubing from bottom-hole to the surface during vertical flow.

Empirical correlations have been developed to predict pressure losses in the tubing for a wide variety of vertical flow condition.

From the wellbore up all the way to the separator, analyzes the performance of the wells need to establish a relationship between the diameters of the pipes, the pressure at the bottom and the wellhead, fluid properties and the flow of production. This relationship is known as the common

name of "Vertical Lift Performance (VLP)" or "Tubing Performance Relationship (TPR)"

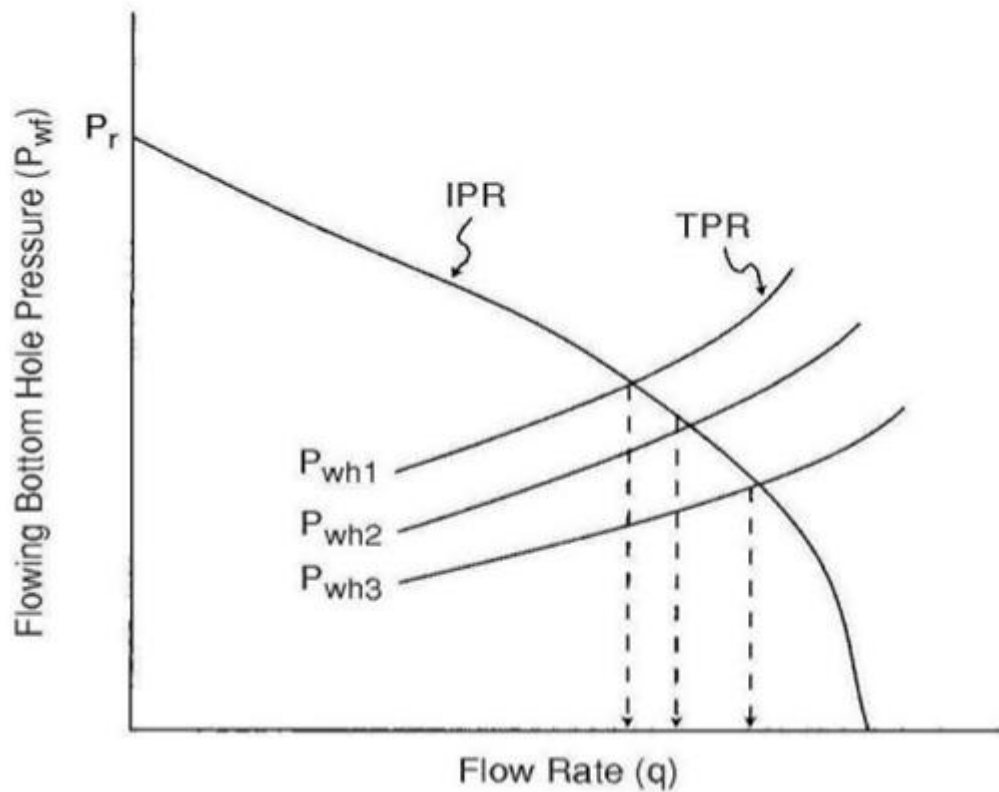


Figure 8: TPR curves for different wellhead pressures

1.2.5.2 Outflow performance sensitivity

The outflow performance is sensitive to:

- Tubing sizing
- water cut
- GOR or injected lift gas
- Size of the sssv (sub-surface-safety valve)
- Choke size
- Wellhead back pressure

1.2.5.3 Multiphase flow theory, patterns and map

1.2.5.4 Operating point.

The interception of IPR curve and VLP curve well zone. VLP: choose an optimal diameter because big size increases hydrostatic pressure losses, and small diameter increases friction pressure losses. Very small Tubing diameter it reduces the capacity production of the well, once installed, it is easy to reduce well diameter than increasing it and effect of GOR rising.

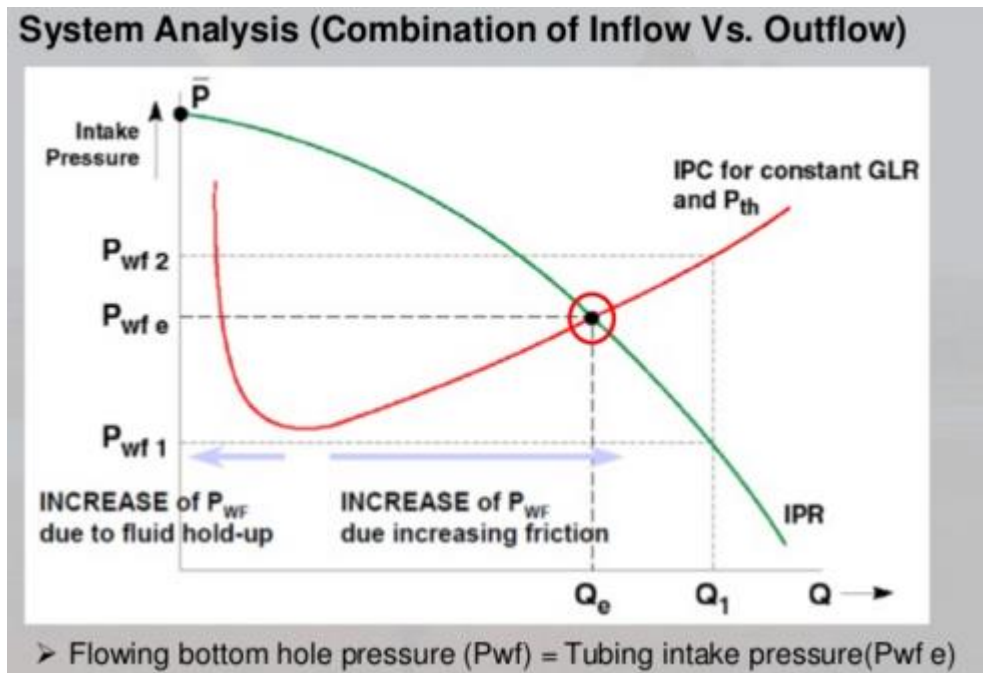


Figure 9 Operating point

3.2.6 Artificial lift

3.2.6.1 Introduction

This module will introduce the topic of artificial lift - a production engineering topic of increasing importance in field development. The reasons leading to this increasing importance in the field development process will be reviewed. The main factors influencing the selection of the most important artificial lift techniques will be highlighted.

A brief description will then be given of all the common artificial lift techniques (rod pumps, electric submersible pumps, progressive cavity pumps and hydraulic pumps) apart from gas lift.

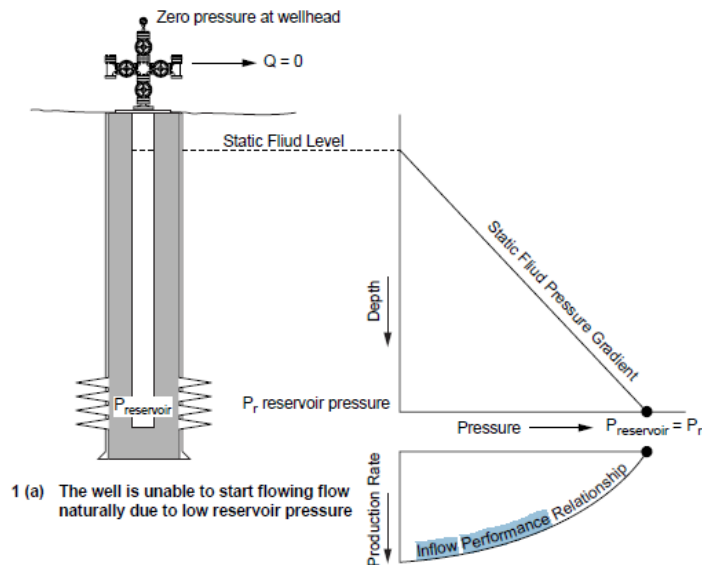
Hydrocarbons will normally flow to the surface under natural flow when the discovery well is completed in a virgin reservoir. The fluid production resulting from reservoir development will normally lead to a reduction in the reservoir pressure, increase in the fraction of water being produced together with a corresponding decrease in the produced gas fraction. All these factors reduce, or may even stop, the flow of fluids from the well. The remedy is to include within the well completion some form of artificial lift. Artificial lift adds energy to the well fluid which, when added to the available energy provided "for free" by the reservoir itself, allows the well to flow at a (hopefully economic) production rate. It has been estimated that in 1994 there was a world inventory of more than 900,000 producing wells. Only 7% of these flowed naturally while the remaining 93% required some form of artificial lift. The average production per well was less than 70 bpd.

THE NEED FOR ARTIFICIAL LIFT

Artificial lift is required when a well will no longer flow or when the production rate is too low to be economic. Figure 1(a) illustrates such a situation - the reservoir pressure is so low that the static fluid level is below the wellhead. Question: Is it possible for this well to flow naturally under and conditions.

Figure 1
Artificial lift fundamentals

Figure 1a
The well is unable to initiate natural flow.



Answer: Yes: If the well productivity Index is sufficiently high and the produced fluid contains enough gas that the flowing fluid pressure gradient gives a positive wellhead pressure. But, the well has to be "kicked off" (started flowing) by swabbing or other techniques.

Figure 1(b) shows how installation of a pump a small distance below the static fluid level allows a limited drawdown ($\Delta p'$) to be created. The well now starts to flow at rate q . N.B. the static and flowing pressure gradients in figures 1(a) & 1(b) are similar since frictional pressure losses in the tubing are small at this low flow rate.

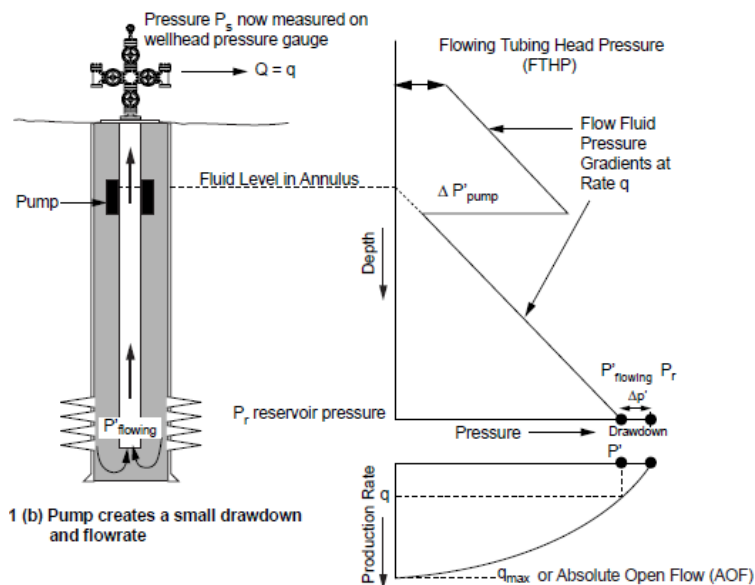
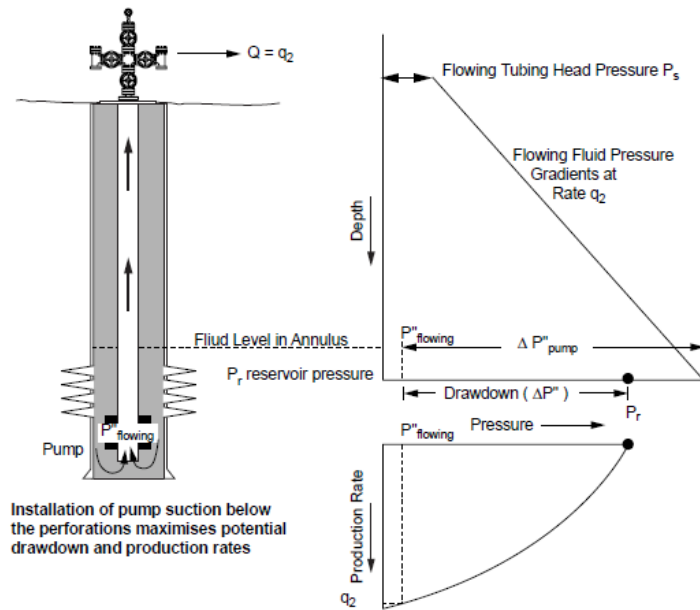


Figure 1b
Pump creates a small drawdown and flow rate

It can be readily seen that the same production rate will occur when the pump is relocated to the bottom of the tubing, provided the pressure drop across the pump, and hence the drawdown, remains the same. The advantage of placing the pump near the perforations is that the maximum potential production can now be achieved {figure 1(c)} by imposing a large drawdown ($\Delta P''$) on the formation and "pumping the well off" by producing the well at q is slightly smaller than the AOF.

Figure 1c
Installation of pump suction below the perforations maximises potential drawdown and production rates

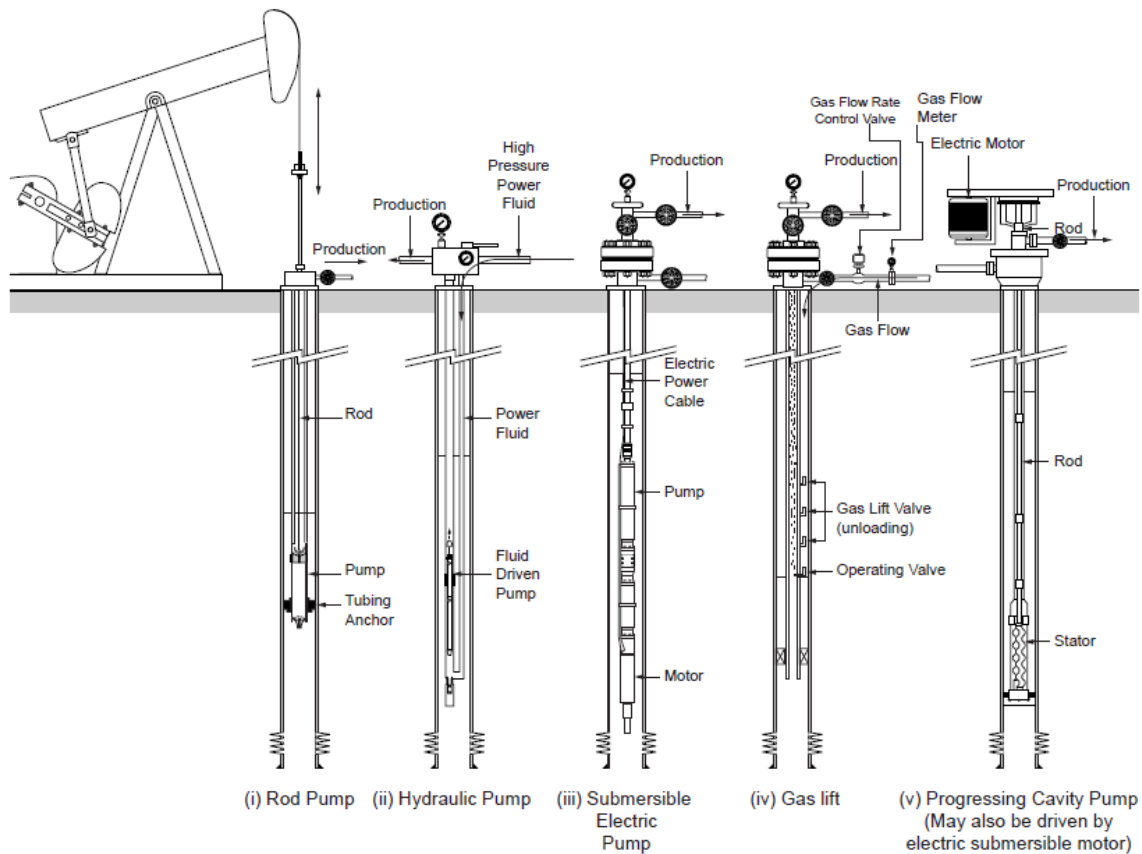


3.2.6.2 Artificial lift forms

Schlumberger book "The concept of artificial lift is needed when reservoir pressures do not sustain acceptable flow rates or there is no fluid flow at all.

Lift process transfers energy down-hole or decreases fluid density in the wellbore to reduce hydrostatic pressure on formations. "

The most popular forms of artificial lift are illustrated in figure 2. They are:



Rod Pumps -



Figure 10: Sucker Rod Pump

Heriot Watt-University "A down-hole plunger is moved up and down by a rod connected to an engine at the surface. The plunger movement displaces produced fluid into the tubing via a pump consisting of suitably arranged travelling and

standing valves mounted in a pump barrel.

The sucker rods, typically 25 ft long, are circular steel rods with diameters between 0.5 in and 1.125 in, in increments of 0.125 in. A threaded male connection or pin is machined at each end of the rod. The two rods can be joined together by use of a double box coupling (Figure 12). Square flats are machined near the pins and at the centre of the coupling to provide a grip for a wrench to allow the rods and couplings to be screwed together. The sucker rods are subjected to continuous fatigue when the pump is in operation. The weight of the rod string is one component of this fatigue load - it can be minimised by using a tapered sucker rod string. This involves installing lighter, smaller diameter rods lower down in the well where the load they have to support (weight of rods and fluid in the tubing string) is less than at the top of the well.

"

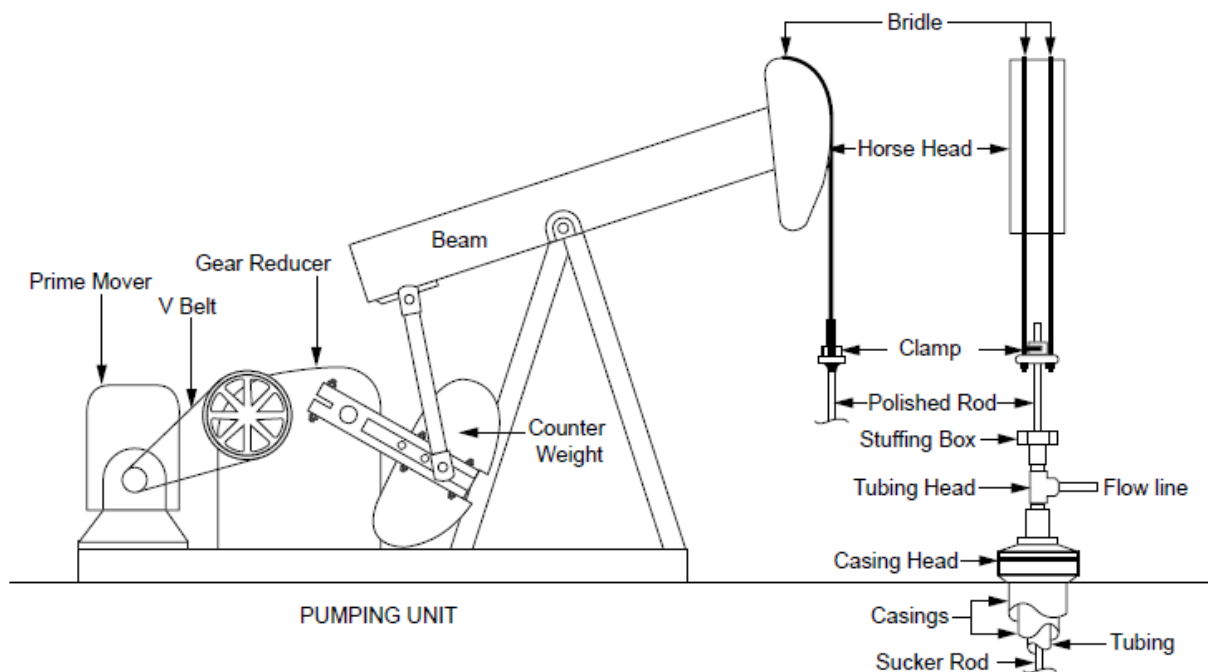


Figure 11: The surface equipment for a Rod Pump

(ii) Hydraulic Pumps

Heriot Watt-University " Hydraulic pumps use a high pressure power fluid pumped from the surface (Figure 37)

which:

- drives a down-hole, positive displacement pump. Figure 38 shown how the flow of power fluid through the upper engine unit is translated into a flow of high pressure produced fluid during both the "UP" and "DOWN" strokes.

(ii) powers a centrifugal or turbine pump (see Section 2.8.3).

(iii) creates a reduced pressure by passage through a venturi or nozzle (Figure 39) where pressure energy is converted into velocity. This high velocity/low pressure flow of the power fluid commingles with the production flow in the throat of the pump. A diffuser then reduces the velocity, increasing the fluid pressure and allowing the combined fluids to flow to surface.

The power fluid consists of oil or production water (the large oil inventory in the surface power fluid system makes oil accounting difficult once high water cuts are being produced). The power fluid is supplied to the downhole equipment via a separate injection tubing. The majority of installations commingle the exhaust fluid

with the production fluid {an “open system” Figure 40(a)}. If difficulties or high costs are encountered in preparing power fluid of the required quality from the production fluid, then a “closed system” may be installed in which the power fluid returns to the surface via a (third) separate tubing {Figure 40(b)}. This option is not available with a venturi pump. The completion design may also allow gas to be vented to surface via the casing/tubing annulus.

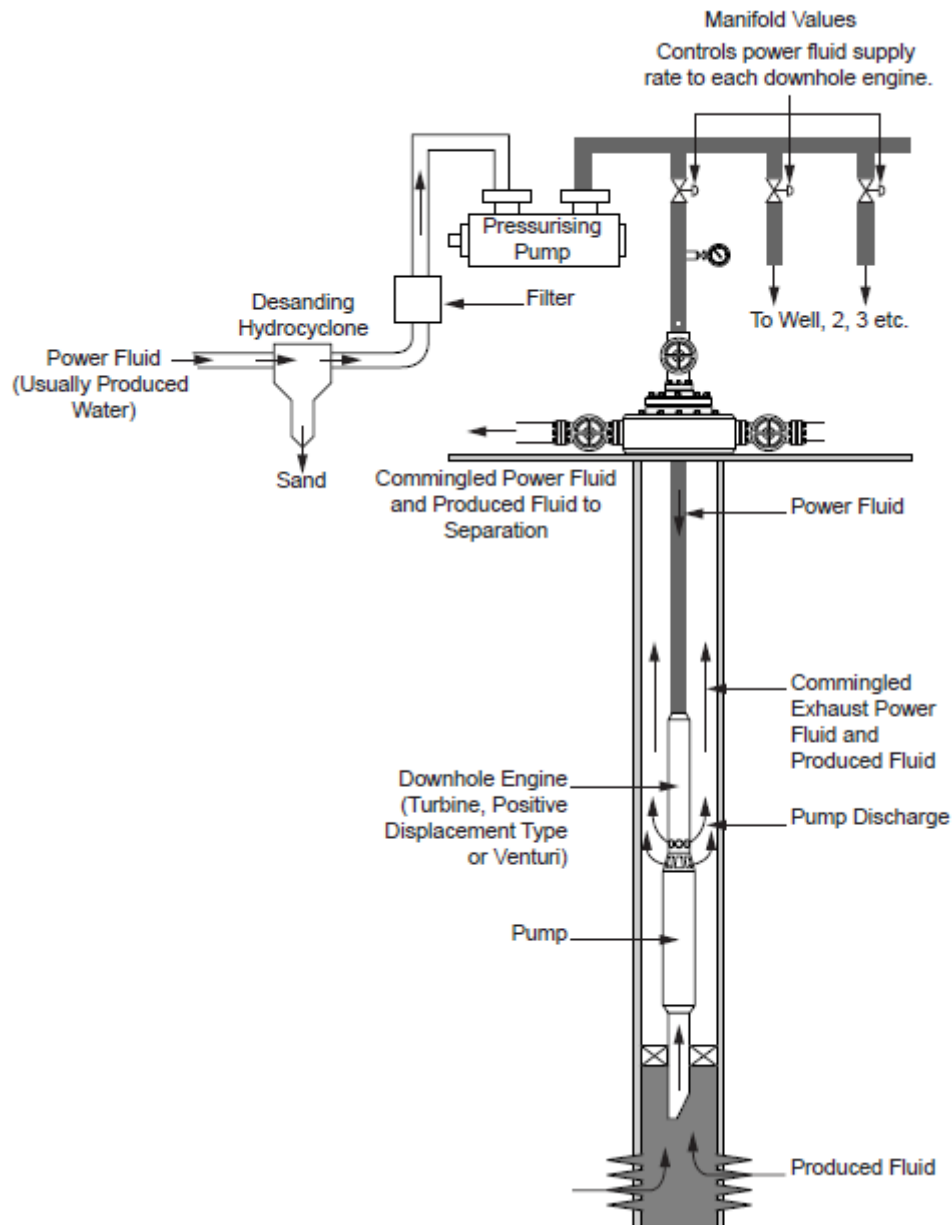


Figure 12: Principles of hydraulic lift operation. Turbine pump is illustrated.

use a high pressure power fluid to:

- (a) drive a down-hole turbine pump or
- (b) flow through a venturi or jet, creating a low pressure area which produces an increased drawdown and inflow from the reservoir.

Islam report:"

Hydraulic pump: this technology used a fluid(usually light oil or water) that is injected from the surface to operate a down-hole pump, forcing water or light oil against the piston and causing pressure to lift the fluid up. A Jet Pump is a type of hydraulic pumps.

As shown in the figure 12, the fluid is injected into the pump body and a small-diameter nozzle, where it becomes a low pressure, high velocity jet. Formation fluid is mixed with the injected fluid and then goes into an expanding-diameter diffuser.

this reduces the velocity while causing the pressure to increase until that is capable of lifting it to the surface. "

(iii) Electric Submersible Pump

Heriot Watt-University " (ESP) employs a downhole centrifugal pump driven by a three phase, electric motor supplied with electric power via a cable run from the surface on the outside of the tubing.

Electric Submersible Pumps (ESP's) are a versatile form of artificial lift with pumps ranging from 150 to 60,000 bfpd in operation. A typical low pressure well that is being artificially lifted using an ESP system is illustrated in figure 17. The functions of the various components are summarized as follows:"

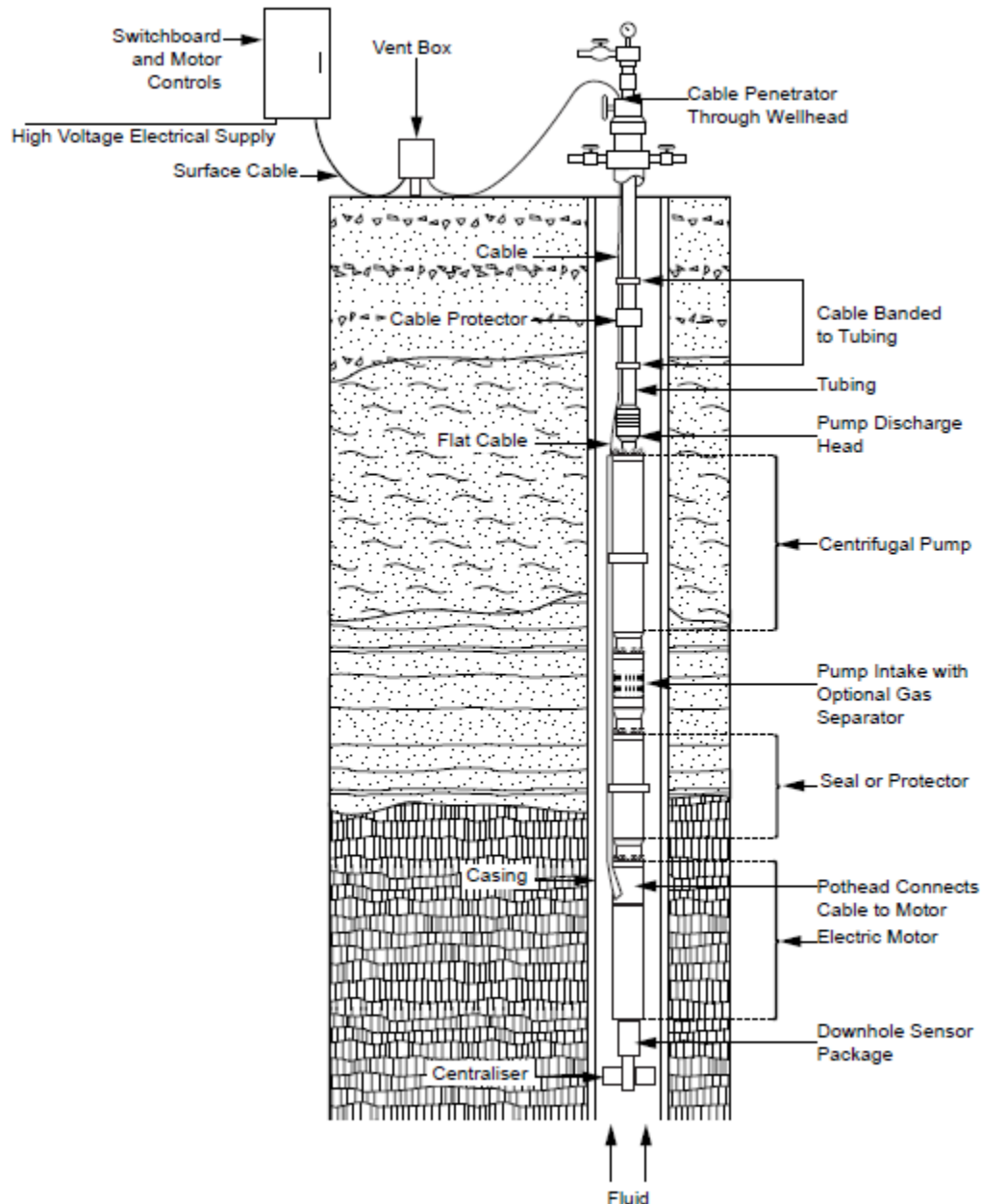


Figure 13: A well completed with artificial lift using an electric submersible centrifugal pump

Islam report: "Electrical submersible pump typically called ESP consists on a down-hole centrifugal pump below the level of the reservoir fluids driven by a submersible electrical motor, which is connected to a power source at the surface. This technology employs spinning impellers on the shaft to put pressure on the surrounding fluids and forcing them to the surface. It can lift more than 25000 barrels of fluid per day"

(iv) **Gas Lift**

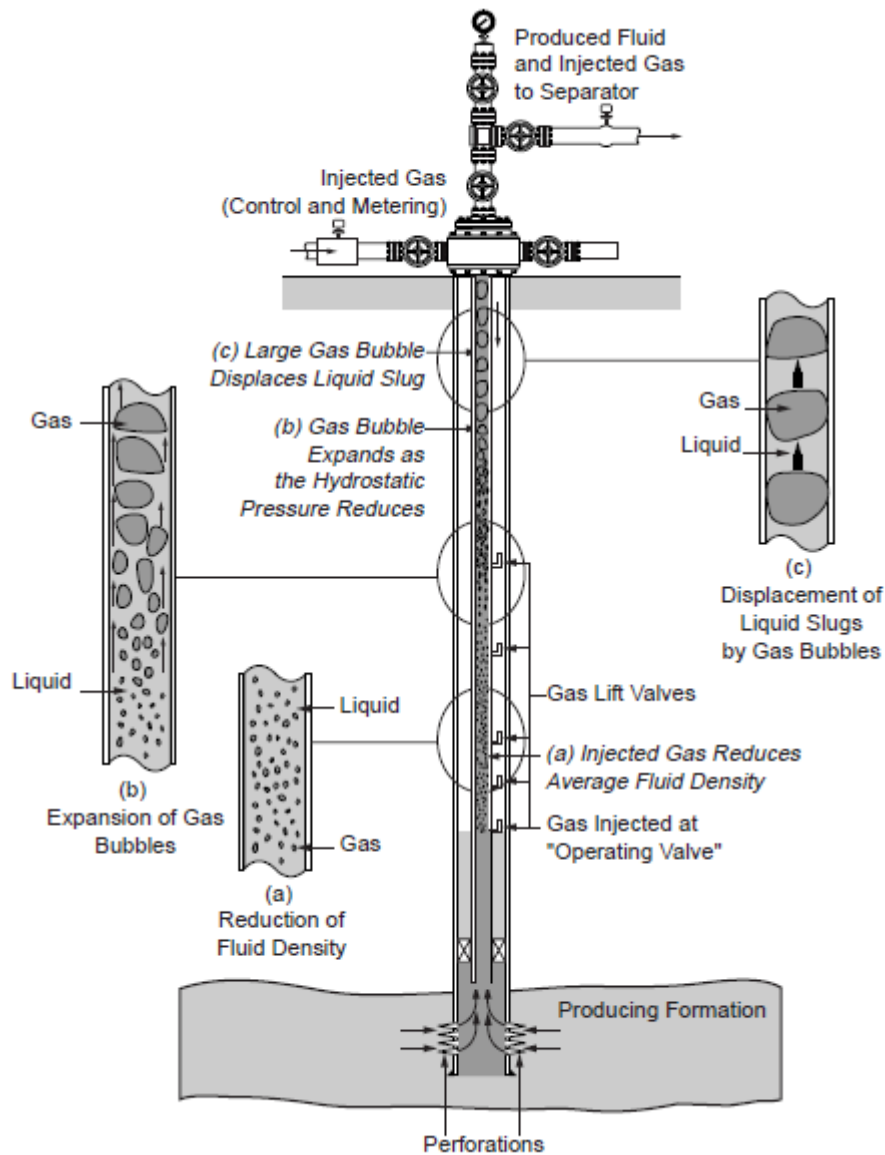
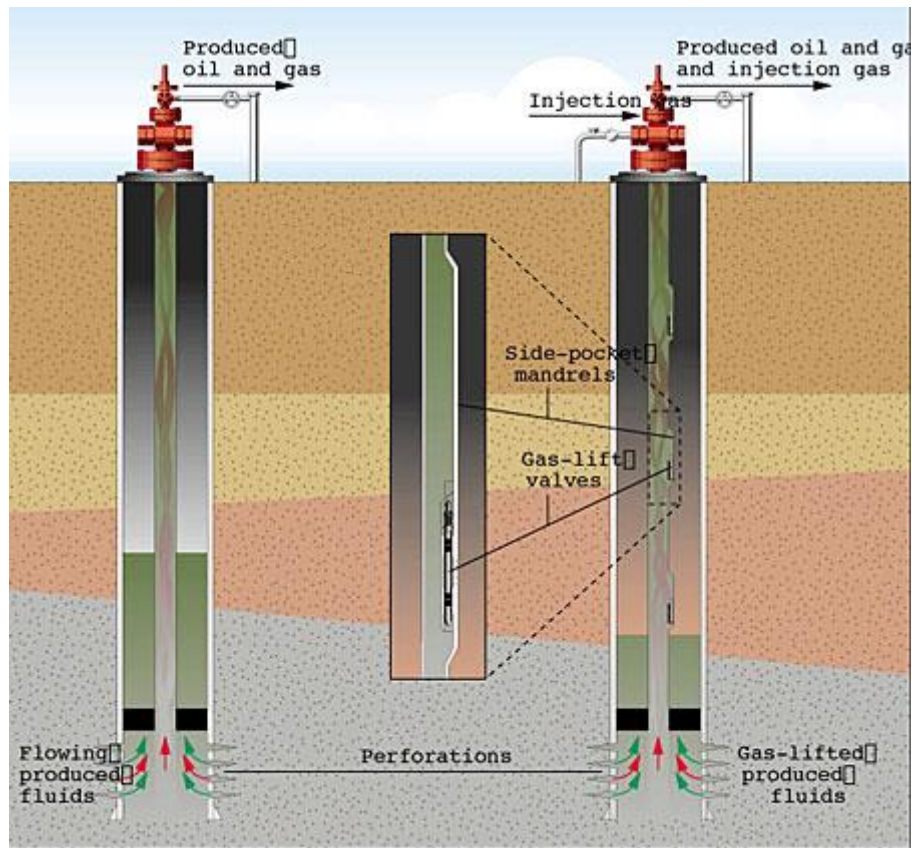


Figure 14: Well completion with gas lift

involves the supply of high pressure gas to the casing/tubing annulus and its injection into the tubing deep in the well. The increased gas content of the produced fluid reduces the average flowing density of the fluids in the tubing, hence increasing the formation drawdown and the well inflow rate.

Slumberger book"



Cross section of wells employing gas lift. Courtesy: Schlumberger.

Gas lift uses additional high pressure gas to supplement formation gas. Produced fluids are lifted by reducing fluid density in wellbore to lighten the hydrostatic column, or back pressure, load on formation.

Application of gas lift :

- to enable wells that will not flow naturally to produce
- to increase production rates in flowing wells
- to unload a well that will later flow naturally
- to remove or unload fluid in gas wells
- to back flow salt water disposal wells
- to lift aquifer wells
- can be intermittent or continuous

Islam report:" For this technique , high-pressure gas is injected into the well to establish pressure. A natural form of gas lift is used in many times for wells flowing without artificial lift. Concerning this artificial lift mean, the injected gas reduces the down-hole pressure by decreasing the fluid viscosity which make the fluid flow more easily to the surface.

The compressed gas is injected down the casing tubing annulus, penetrating the well from different gas-lift valves creating bubbles that lighten the fluids.

There are two types of gas lift used in the oil industry: continuous flow gas lift and intermittent gas lift."

(v) Progressing Cavity Pump

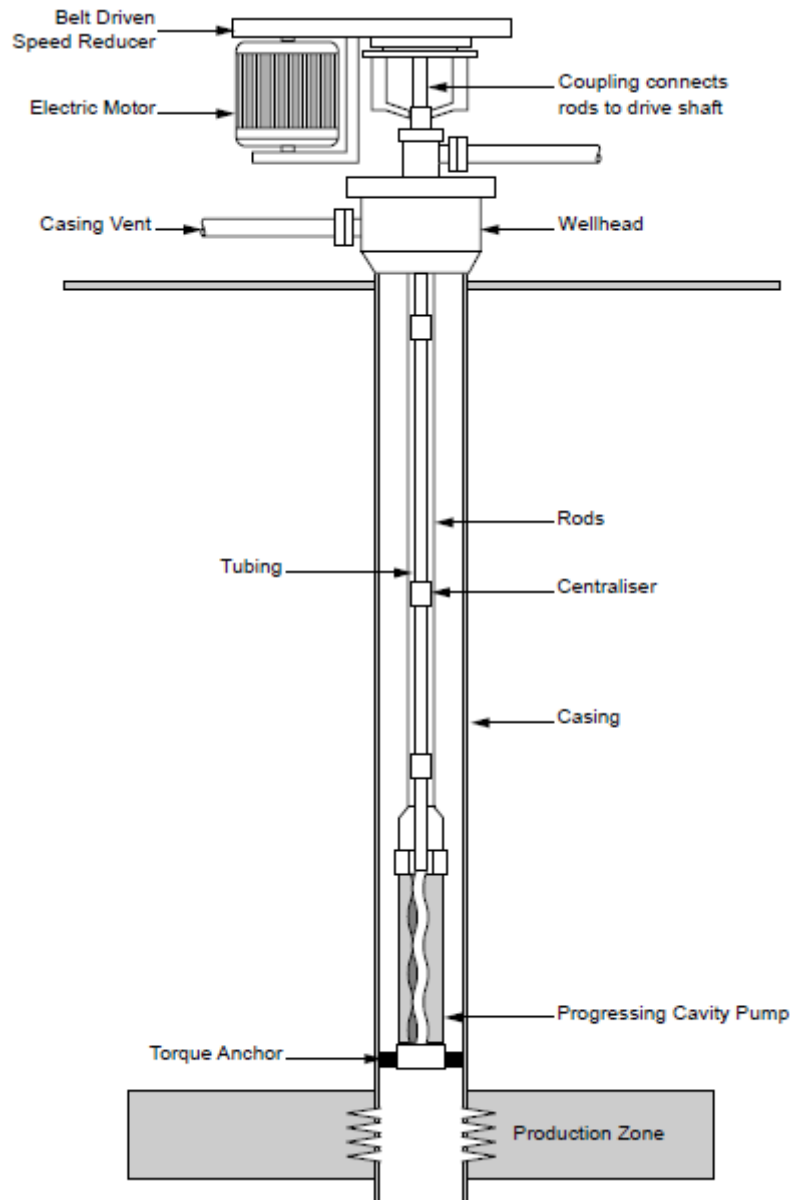
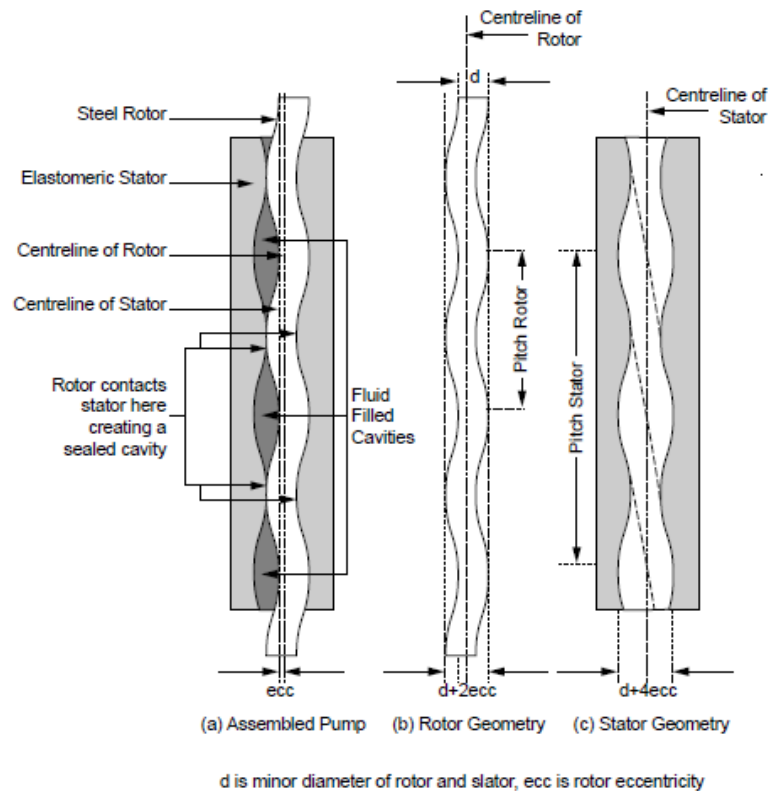


Figure 15: A well completed with AL using a Progressive Cavity Pump

Figure 14 illustrates the main components of a PCP. A steel shaft rotor of diameter d has been formed into a helix {Figure 44(a)}. The rotor is rotated inside an elastomeric pump body or stator, which has been molded in the form of a double helix with a pitch of the same diameter and exactly twice the length of the pitch given to the rotor {Figure 44(b)}. Figure 44(c) shows that, when assembled, the centre line of the rotor and the stator are slightly offset, creating a series of fluid filled cavities along the length of the pump. Figure 45 is a perspective view of Figure 44(c), which helps explain how the interference fit between the rotor and stator creates two chains of spiral (fluid filled) cavities.

(PCP) employs a helical, metal rotor rotating inside an elastomeric, double helical stator. The rotating action is supplied by down-hole electric motor or by rotating rods.



Slumberger book" Standard PC Pump-Topdrive

- Down-hole pump components:

-Rotor

-Stator

- Sucker Rod string
- Surface drive head
- Accessories:

-Torque anchor.

-Rod Protectors/ centralizers

PC Pump applications:

- heavy and viscous oil
- Production of solids-laden fluids
- medium to sweet crude
- coal bed methane / gas well de-watering
- Urban areas
- agricultural areas

...

"

3.2.6.3 Advantages, disadvantages and of different artificial lift technologies:

The “Advantages and Disadvantages of the Major Artificial Lift Methods” are listed and compared in Tables

Rod Pumps	Electric Submersible Pump	Venturi Hydraulic Pump	Gas Lift	Progressing Cavity Pump
Simple, basic design Unit easily changed Simple to operate Can achieve low BHFP Can lift high temperature, viscous oils Pump off control	Extremely high volume lift using up to 1,000 kw motors Unobtrusive surface location Downhole telemetry available Tolerant high well elevation / doglegs Corrosion / scale treatments possible	High volumes Can use water as power fluid Remote power source Tolerant high well deviation / doglegs	Solids tolerant Large volumes in high PI wells Simple maintenance Unobtrusive surface location / remote power source Tolerant high well deviation / doglegs Tolerant high GOR reservoir fluids Wireline maintenance	Solids and viscous crude tolerant Energy efficient Unobtrusive surface location with downhole motor

*Table 2
Advantages of major artificial lift methods*

Rod Pumps	Electric Submersible Pump	Venturi Hydraulic Pump	Gas Lift	Progressing Cavity Pump
Friction in crooked / holes Pump wear with solids production (sand, wax etc.) Free gas reduces pump efficiency Obtrusive in urban areas Downhole corrosion inhibition difficult Heavy equipment for offshore use	Not suitable for shallow, low volume wells Full workover required to change pump Cable susceptible to damage during installation with tubing Cable deteriorates at high temperatures Gas and solids intolerant Increased production casing size often required	High surface pressures Sensitive to change in surface flowline pressure Free gas reduces pump efficiency Power oil systems hazardous High minimum FBHP. Abandonment pressure may not be reached	Lift gas may not be available Not suitable for viscous crude oil or emulsions Susceptible to gas freezing / hydrates at low temperatures High minimum FBHP. Abandonment pressure may not be reached Casing must withstand lift gas pressure	Elastanes swell in some crude oils Pump off control difficult Problems with rotating rods (windup and after spin) increase with depth

*Table 3
Disadvantages of major artificial lift methods*

4. Properties of reservoir fluids and phase behavior

4.1. Multiphase flow theory, patterns and map

- Flow theory

The three components of the equation for predicting pressure losses are: elevation or static components, friction component, acceleration component.

$$\Delta P_{total} = \text{Elevation hydrostatic}) + \text{Friction} + \text{Acceleration}$$

- Flow patterns

vertical flow

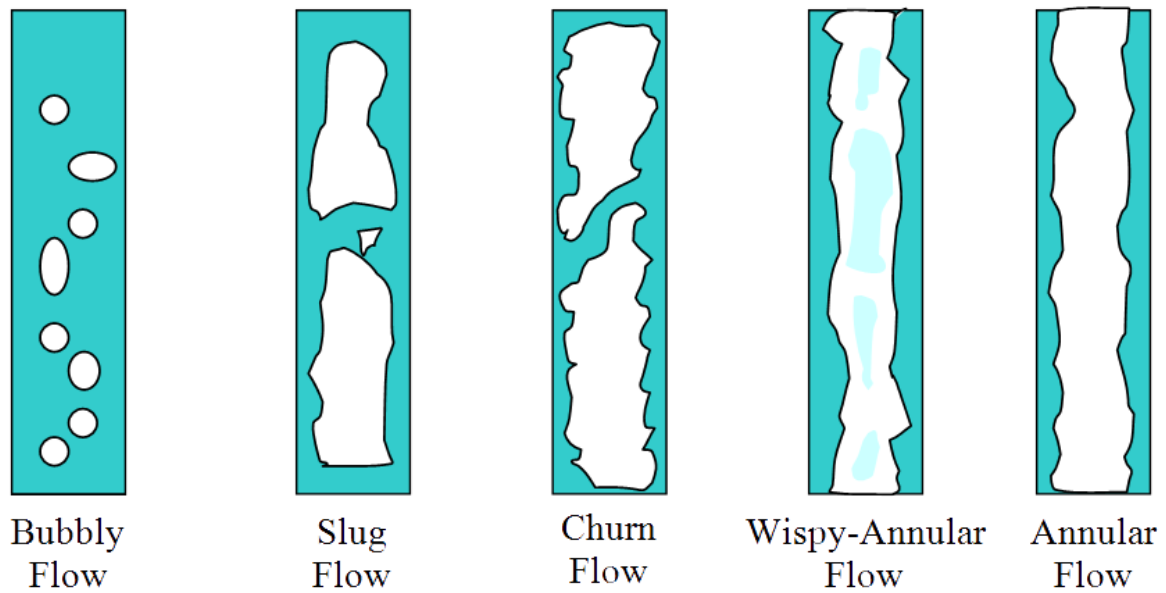


Figure 16: Vertical flow patterns

Bubble flow: Numerous yet discrete gas bubbles are dispersed in the continuous liquid phase.

slug flow: larger bubbles are formed with sizes similar to the tubing diameter. They are separated from one another by slugs of liquids.

churn flow: Higher velocities change the aspect of the flow; it becomes very unstable which threatens to damage the pipe.

wispy-annular flow: when the flow rates gets even higher the small droplets form clouds of liquid in the center gaseous core.

annular flow: Gas velocity exceeds the liquid's velocity. The liquid travels then in the tube as thin film on the wall as the gas flows as a continuous phase.

horizontal flow

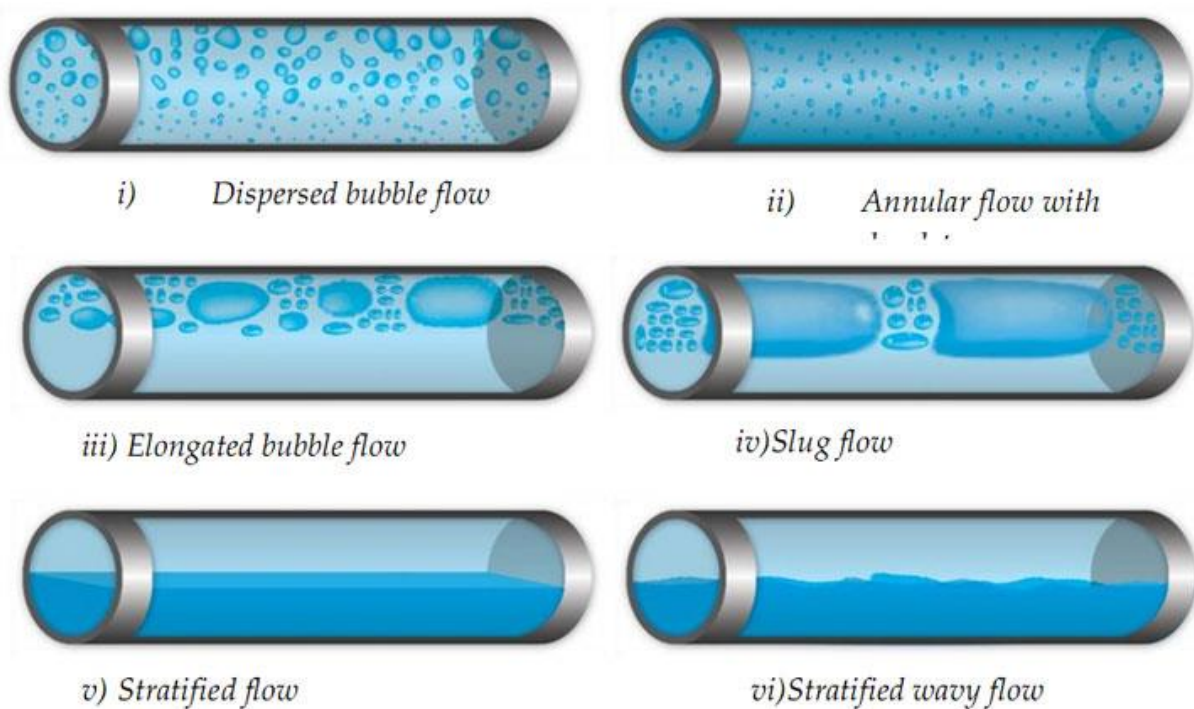


Figure 17: Horizontal flow patterns

Bubble flow: Both gas and liquid move with the same velocity as the gas is dispersed as bubbles that tend to accumulate at the top of the tubing.

slug flow: At higher gas velocities this regime occurs with its bigger elongated bubbles and large vibrations caused by the liquid slugs between the bubbles.

annular flow: At even greater gas velocities, the liquid forms a continuous annular film that gets thicker at the bottom of the tube bubble point.

Stratified flow: At low liquid and gas velocities, the two phases are completely separated. The liquid goes to the bottom as the gas goes to the top.

Wavy flow: Increasing the fluids velocity in a stratified flow, waves are formed

4.2.Bubble point.

As defined by **Shlumberger Glossary** the bubble point means" The pressure and temperature conditions at which the first bubble of gas comes out of solution in oil."

Below the bubble point the solution of oil is saturated with gas, meaning that the oil contains the maximum amount of gas that could it holds. So as the pressure drops along the way from the bottom-hole to the well-head the gas will be separated from the solution as form of bubbles and the oil will be unsaturated.

4.3.GOR(Gas Oil Ratio)

The oil gas ratio is the ratio between the volume of gas dissolved(measured at standard conditions) and the volume of oil at standard conditions.

$$R_s = \frac{V_g(P_{atm}, 60^\circ F)}{V_o(P_{atm}, 60^\circ F)}$$

4.4.FVF(Formation Volume Factor)

4.4.1. Formation Volume Factor (oil)

AS defined by **Perowiki** " The oil formation volume factor (FVF) relates the volume of oil at stock-tank conditions to the volume of oil at elevated pressure and temperature in the reservoir. Values typically range from approximately 1.0 bbl/STB for crude oil systems containing little or no solution gas to nearly 3.0 bbl/STB for highly volatile oils."

$$B_o = \frac{\text{Volume of oil in reservoir, (Pr, Tr condition)}}{\text{Volume of stock tank oil in standard conditions}}$$

4.4.2. Formation Volume Factor (gas)

The formation volume factor of gas is defined by **Perowiki** as " the ratio of the volume of gas at the reservoir temperature and pressure to the volume at the standard or surface temperature and pressure (p_s and T_s). It is given the symbol B_g and is often expressed in either ((rcf) cubic feet of reservoir volume per(sc) standard cubic foot of gas) or(barrels of reservoir volume per standard cubic foot of gas)".

$$B_g = \frac{V_R}{V_{sc}} = \left(\frac{znRT}{p} \right) \left(\frac{P_{sc}}{z_{sc}nRT_{sc}} \right) = \frac{P_{sc}zT}{T_{sc}P}$$

$$\text{where } \frac{P_{sc}}{T_{sc}} = \frac{14.696(Psi)}{519.67(^{\circ}R)} = 0.0282793$$

$$\text{which implies } B_g = 0.0282793 \frac{zT}{P} \text{ (rcf/scf)}$$

where:

V_R : Reservoir volume

V_{sc} : Volume in standard condition

T : Reservoir Temperature

T_{sc} : Temperature in standard condition

P_{sc} : Pressure in standard condition

n : Number of mole

z : Compressibility factor (gas deviation factor)

z_{sc} : Compressibility factor at standard condition

R : gas-law constant

The n divides out here because both volumes refer to the same quantity of mass.

4.5. Water cut (WC)

Water cut is the ratio of water produced to the total volume of fluids produced: oil + water, both volumes measured in standard conditions. It is expressed as a fraction in percent:

$$\%Wc = \frac{V_w}{V_t} \times 100$$

4.6. Oil density

According to "**Petrowiki**" The density of a reservoir gas is defined as the mass of the gas divided by its reservoir volume, so it can also be derived and calculated from the real gas law:

$$\rho_g = \frac{m_g}{V_R} = \frac{nM_g}{znRT/P} = \frac{nM_{air}\gamma_g}{znRT/P} = \frac{28.967P\gamma_g}{zRT}$$

Where

V_R : Reservoir volume

T : Reservoir Temperature

n : Number of mole

z : Compressibility factor (gas deviation factor)

R : gas-law constant

m_g : Reservoir volume

M_g : average molecular weight of gas mixture

M_{air} : molecular weight of air

ρ_g : density of gas

γ_g : Specific gravity of gas

m_g : mass of gas

5. Conclusion

Chapter3: Electrical Submersible Pump(ESP)

1. Introduction

1.1.General history of ESP

Source: ESP manuel pdf"Gabor Tackas"

Unlike most of the other artificial lift methods such as gas lifting or suckerrod pumping, whose invention cannot be attributed to any person or any definite time, electrical submersible pumping was invented and developed by a Russian named Arutunoff in the late 1910s [2].

In 1911, Arutunoff started the company Russian Electrical Dynamo of Arutunoff (its acronym REDA still being known all over the world) and developed the first electric motor that could be operated submersed in an oil well. To acquire funding for the development of his ideas, Arutunoff first emigrated to Germany in 1919, and then finally settled in the USA in 1923. The US Patent he received on the electrical submersible pump [3] was issued in 1926 and covered the principal features of this new artificial lift method. The first ESP installation was successfully operated in the El Dorado field in Kansas in 1926. Arutunoff moved to Bartlesville, Oklahoma, in 1928 where he started the Bart Manufacturing Co., later reorganized as REDA Pump Co. in 1930. The first ESP units were driven by three-phase two-pole electric induction

motors of 53

=8 in or 71

=4 in OD. The biggest motor was about 20 ft

long and developed 105 HP. Directly above the motor a seal unit was attached whose main task was to prevent the leakage of well fluids into the motor. On top of the seal unit, a multistage centrifugal pump lifted well fluids to the surface. The complete ESP unit (motor, seal and pump) was run into the well on the bottom of the tubing string, electricity being supplied from the surface to the motor by a special three-conductor cable. Even today, these are the main components of electrical submersible pumping installations. After more than 80 years of operation, the company established by Arutunoff, who alone received 90 patents related to submersible equipment, is still one of the leading suppliers of ESP equipment to the world's petroleum industry.

From its conception, ESP units have excelled in lifting much greater liquid rates than most of the other types of artificial lift and found their best use in high rate on- and offshore applications. It is believed that today approximately 10% of the world's oil supply is produced with submersible pumping installations.

During its long history, ESP equipment underwent a continuous improvement. The first breakthrough came in the early 1950s when seal units (a.k.a. protectors) with mechanical seals on their shafts considerably increased ESP run lives because they provided a much better protection against leakage of well fluids into the motor. Production of gassy wells was always a problem and the use of simple gravitational (reverse-flow)

gas separators did not solve the problem completely until the first rotary gas separator [4] was introduced in the early 1970s. Although the other components of the ESP unit have also evolved, the next revolutionary moment came when the first variable speed ESP unit was installed in August 1977 [5]. The variable speed drive (VSD) changes the frequency of the electric current driving the ESP motor and thus considerably modifies the head performance of the submersible pump. By properly setting the driving frequency, a very basic limitation of ESP units can be eliminated and the lifting capacity of the submersible pump can easily be modified to match the inflow performance of the well. Without a VSD unit, in wells with unknown liquid production capacities the ESP unit has to be exchanged with a unit better fitting the inflow to the well, which usually involves a costly workover operation.

Running and pulling of conventional ESP units involves the use of heavy workover units because the tubing string has to be moved into or out of the well. Reduction of the high workover costs can be accomplished if the ESP unit is run on a wire rope of the right mechanical strength. Cable suspended units, first appearing in the oil field in the late 1970s, became very popular for their advantageous features, especially in the offshore environment [6]. Similar advantages can be reached with coiled tubing (CT) conveyed ESP units, first installed in Alaskan fields in 1992 [7]."

1.2.General overview of ESP

AI- ESP pdf"

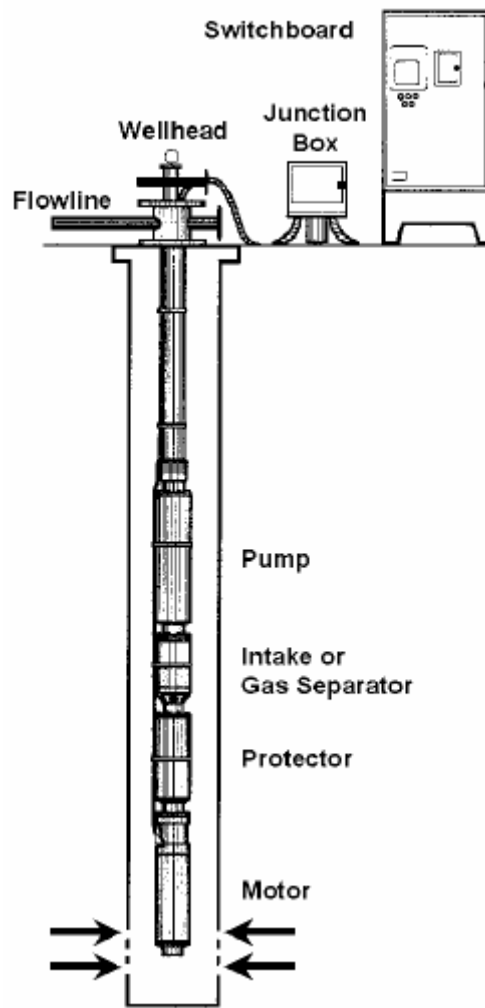


Figure 18: Conventional ESP installation

means of producing liquid from oil and water wells. The classical or “**conventional**” installation is illustrated in **Fig X** where the ESP unit is run on the tubing string and is submerged in well fluids. The electric submersible **motor** is at the bottom of the unit and is cooled by the well stream passing by its perimeter. It is connected to the **protector** (a.k.a. seal) section that provides many crucial functions for the safe operation of the unit. On top of the protector a pump intake or gas separator is situated which allows well fluids to enter the centrifugal pump and, at the same time, can remove low quantities of free gas from the well-stream. Liquid is lifted to the surface by the **multistage centrifugal pump**, the heart of the ESP system.

The submersible motor is supplied with three-phase AC current via an **electric cable** running from the surface all along the tubing string. Produced fluids flow thorough the tubing string to the surface, where a special **wellhead** ensures feeding of the electric cable into the well. Surface equipment includes a **junction box** where down-hole and surface electric cables are joined and a control unit (called **switchboard**) that provides measurement and control functions. The ESP unit receives AC electricity from a set of **transformers** (not

shown) which supply the required voltage by stepping up or down the voltage available from the surface electric network.

The above type of installation has stood the test of time and is frequently used even today. The most important features making it a **conventional ESP installation** are:

- Only **liquid** enters the centrifugal pump providing ideal conditions for the pump. The well must produce only a low amount of free gas at the pump suction that can easily be removed by the gas separator.
- The viscosity of the produced liquid is low, approaching the viscosity of water.
- The ESP motor is supplied with an AC current of a **constant frequency** thus its speed and consequently that of the centrifugal pump are constant.

Although the above conditions are not always met, the conventional ESP installation can be applied in a great variety of field conditions."

2. Basic features

The heart of the ESP unit is the submersible pump and the design and analysis of the whole ESP system cannot be understood without a basic comprehension of the operation of the pump. This is the reason why the description of the system's components has to be started with a thorough analysis of the construction and operation of the centrifugal pumps.

The submersible pumps used in ESP installations are **multistage centrifugal pumps** operating in a vertical position. Although their constructional and operational features underwent a continuous evolution over the years, their basic operational principle remained the same. Produced liquids, after being subjected to great centrifugal forces caused by the high rotational speed of the **impeller**, lose their kinetic energy in the **diffuser** where a conversion of kinetic to pressure energy takes place. This is the main operational mechanism of radial and mixed flow pumps, as detailed in **Chapter Two**.

Fig. 3.2 illustrates the main parts of an ESP pump containing mixedflow stages. The pump shaft is connected to the gas separator or the protector by a mechanical coupling at the bottom of the pump. Well fluids enter the pump through an **intake screen** and are lifted by the pump stages. Other parts include the radial **bearings** (bushings) distributed along the length of the shaft providing radial support to the pump shaft turning at high rotational speeds. An optional **thrust bearing** takes up part of the axial forces arising in the pump but most of those forces are absorbed by the protector's thrust bearing. The liquid producing capacity of an ESP pump depends on the following factors:

- the rotational **speed** provided by the electric motor,
- the **diameter** of the impeller,
- the **design** of the impeller (characterized by its specific speed),

- the **number** of stages,
- the actual **head** against which the pump is operating, and
- the thermodynamic properties (density, viscosity, etc.) of the produced fluid.

Conventional ESP installations run on AC power with a constant frequency of 60 Hz or 50 Hz. ESP motors in 60 Hz electrical systems rotate at a speed of about 3,500 RPM, whereas in case of a 50 Hz power supply the motor speed is about 2,900 RPM. For constant speed applications the most important factor is **impeller size** which, of course, is limited by the ID of the well casing. Pumps of bigger sizes can produce greater rates although impeller design also has a great impact on pump capacity. Present-day ESP pumps come in different capacities from a few hundred to around 80,000 bpd of liquid production rate and in outside diameters from around 3" up to 11". Smaller units contain "**pancake**" type impellers with **radial discharge** and are used up to the rates of 1,500 – 3,500 bpd, above which **mixed flow** impellers are used. The **length** of individual ESP pumps is limited to about 20 - 25 ft, for ensuring proper assembly and ease of handling. **Tandem pumps** are made up of several pump sections (up to three) and are used to achieve higher operational heads usually required in deeper wells. This way several hundreds of stages can be run, the maximum number of stages being limited by one or more of the following factors:

- the mechanical strength of the **pump shaft**, usually represented by the shaft's horsepower rating,
- the maximum **burst-pressure rating** of the pump housing, and
- the maximum **allowed axial load** on the unit's main thrust bearing (usually situated in the

protector section). Individual stages in ESP pumps, provided they are of the same impeller design, handle the same liquid volume and develop the same amount of head. The heads in subsequent stages are **additive** so a pump with a given number of stages develops a total head calculated as the product of the total number of stages and the head per stage. This rule allows one to find the number of stages required to develop the total head to overcome the total hydraulic losses, valid at the desired liquid production rate in a well. Since the size of well casing limits the outside diameter of the ESP equipment which can be run, pump selection is heavily restricted by the actual casing size. **Appendix B** lists the main dimensional data of common API tubular. For comparison, **Appendix C** contains the most important parameters (diameters, recommended liquid rate ranges) of submersible pumps available from a leading manufacturer.

3. Pump performance curves

The performance of ESP pumps is characterized by the **pump performance curves** introduced in

Section 2.2.2.3. These are plotted in the function of the pumping rate and represent:

- the **head** developed by the pump,
- the **efficiency** of the pump, and
- the mechanical **power** (brake horsepower) required to drive the pump when pumping water.

These curves are **experimentally** obtained with freshwater under controlled conditions described in **API RP 11S2** [1] at an operating temperature of 60°F. Tests on submersible pumps are made by driving the pump at a **constant** rotational speed, usually 3,500 RPM for 60 Hz service. This is the speed generally assumed as the actual speed of a fully loaded ESP motor having a synchronous speed of 3,600 RPM. Experiments with other fluids or speeds must be corrected to these standard conditions using the **Affinity Laws**. All performance parameters must represent the operation of one or a specified number of pump stages clearly indicated on the chart.

Performance curves of pumps in 50 Hz service are typically published for a fixed rotational speed of 2,917 RPM. As described by **Butlin** [2], this number was chosen based on a misunderstanding of **slip** in electric motors, using the concept of the “percentage slip”. It was erroneously assumed that motor slip is found as a percentage of the synchronous speed. Slip (defined as the difference between the motor’s synchronous and actual speeds), however, depends on motor voltage and torque loading only.

This means that the same motor at the same pump load will have the same amount of slip, independent of the electric frequency. The synchronous speed of ESP motors at 50 Hz service being 3,000 RPM, actual full-load motor speeds must be around 2,900 RPM or even less for small diameter motors.

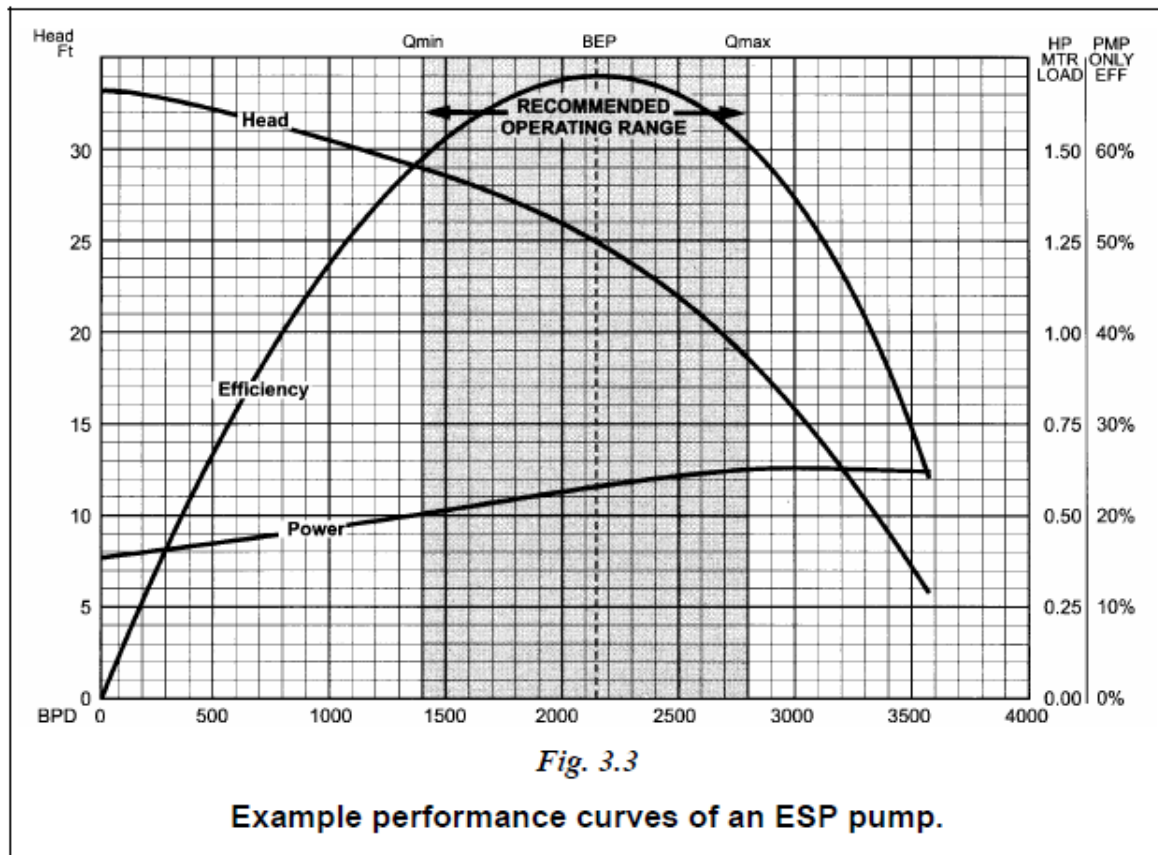
Due to manufacturing tolerances and other factors, a variance in performance as compared to published curves can exist from one pump to another of the same design. The accepted tolerances between published curves and actually tested pump performance are specified in **API RP 11S2** [1]

where the guidelines contained in **Table 3.1** are given. The sample curves shown in **Fig. 3.3** represent the features of one stage of a given pump design. The performance parameters belonging to the **best efficiency point** (BEP) represent the criteria for an

Table 3.1

ESP pump performance data tolerance limits.

Parameter	Tolerance	Applied to
Head	$\pm 5\%$	Recommended range
Flow Rate	$\pm 5\%$	Recommended range
BHP	$\pm 8\%$	Recommended range
Efficiency	- 10%	Best Efficiency Point



optimum utilization of the pump, around which the recommended range of operation is indicated.

The recommended range of pumping rates for any ESP pump is strictly related to the variation of **axial forces** occurring in the pump. As discussed in **Section 2.2.2.5**, static and dynamic axial forces occurring in pump stages are the results of different phenomena and may take the form of down-thrust or up-thrust. **Fig 3.4** presents the relationship between the axial forces and the recommended operating range of the pump; their interaction is detailed in the following.

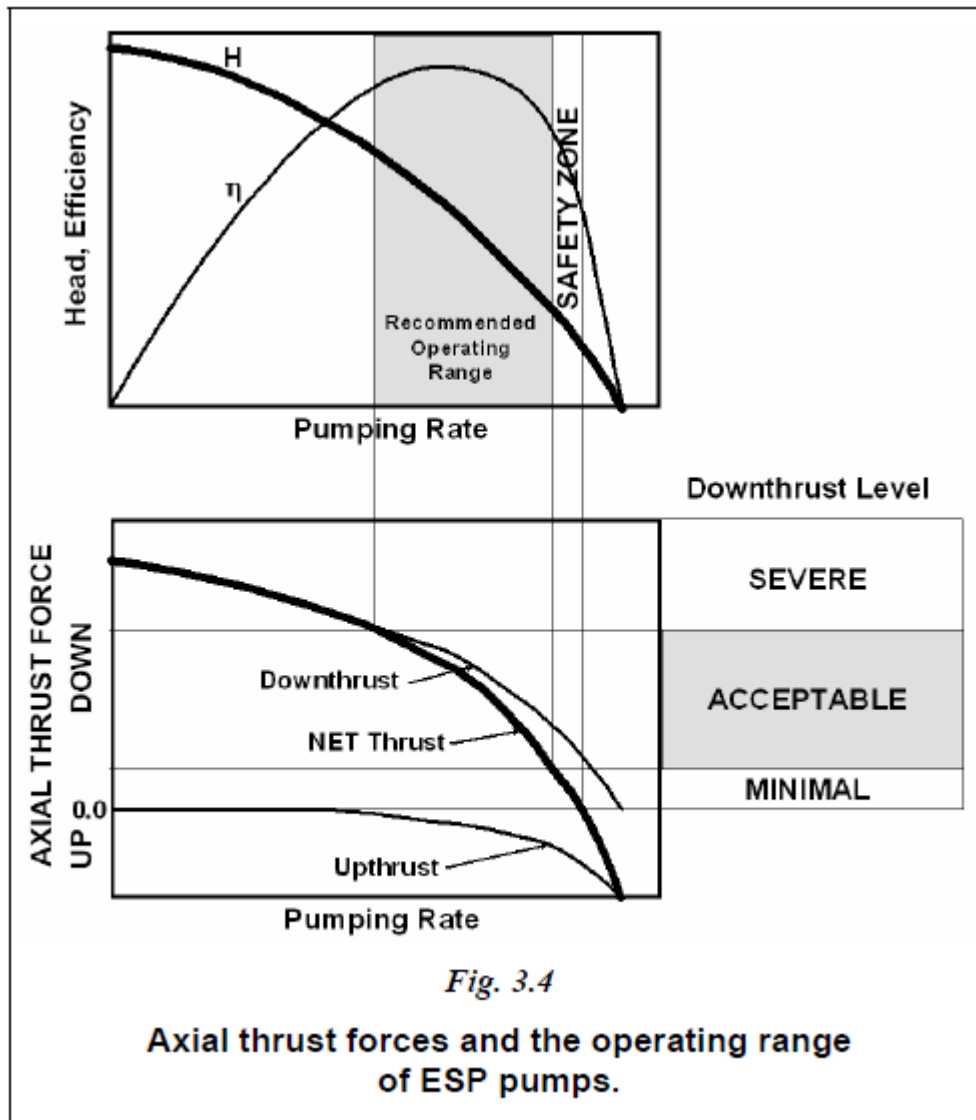
The lower part of **Fig. 3.4** shows schematically the change of **axial forces** in an ESP pump stage in the function of the liquid rate produced by the pump. **Downthrust** is basically determined by the **head** developed since its main component comes from the pump's discharge pressure acting on the top and bottom shrouds of the impeller. Its variation with pumping rate, therefore, follows the shape of the pump's head – rate performance curve, as

shown in the figure. It is at a maximum at shut-in conditions (at a pumping rate of zero), and diminishes to zero where the pump's head decreases to zero. **Upthrust** forces arising in the pump stage, on the other hand, are the result of the change in **inertial** forces and are proportional to the kinetic energy of the liquid pumped. Thus their variation with pumping rate follows a second-order curve.

The sum of the up- and downthrust forces is shown in a bold line representing the **net thrust** arising in the pump stage. As seen, the operation of ESP pumps is dominated by downthrust forces because the net axial force points downward in the largest part of the operating pumping rates.

As discussed in **Section 2.2.2.5**, axial forces developed in ESP pumps must be compensated, otherwise the axial movement of the impellers and the pump shaft leads to mechanical **damage** of the stages. Elimination of such forces is accomplished differently in stages with **fixed** impellers from stages with **floating** impellers.

- In fixed impeller pumps all axial forces are transmitted to the pump shaft and have to be balanced by the main thrust bearing, situated in the protector section of the ESP unit. This solution necessitates the use of thrust bearings of relatively large capacity.
- In floating impeller pumps, on the other hand, most of the axial forces are compensated by frictional forces arising in the up-, and down-thrust **washers** installed on the impellers, requiring smaller-capacity thrust bearings.



In the following discussion, let us assume a **floating** impeller pump and determine the safe operating range of such ESP pumps. Most of the net thrust illustrated in **Fig. 3.4** must be absorbed by the thrust washers in the impeller. Since (because of construction reasons, see **Fig. 2.8**) the load carrying area of the upthrust washer is much smaller than that of the downthrust washer, upthrust is more **dangerous** for the pump's operation than downthrust. To safeguard against the occurrence of upthrust, a minimum of downthrust is assumed and a safety zone in pumping rates is created, as shown in **Fig. 3.4**. As illustrated in the figure, this defines the upper boundary of the pump's applicability in such a way that upthrust conditions are safely avoided.

Since **downthrust** forces are absorbed by the combined effort of the washers and the unit's main thrust bearing situated in the protector, their combined capacity defines the maximum acceptable axial load that can be allowed. The pumping rate belonging to this load defines the minimum recommended pumping rate of the ESP pump. Any given pump, therefore, must be operated in the **recommended** operating range defined by the two boundary rates described previously. Keeping the operation of an ESP pump inside the recommended pumping rate range poses the most severe **restriction** in the application of ESP equipment and should never be forgotten by users.

4. Glossary

SAMIA GHRIBI report"

- Pump erosion: is caused by abrasive solid particles striking the metal surfaces just like in sandblasting
- The wear: is proportional to the solid size roughness and kinetic energy.
- Abrasives: partials propelled at a velocity sufficient to cause cleaning and wearing away a surface

The severity of abrasive damages increases in the following order:

-erosion in impellers and diffusers

-axial wear in thrust bearing, up and down thrust washers in floater pumps

-radial wear in radial bearings

Up thrust: the recirculation forces are greater on the lower shroud of the impeller moving it up

down thrust: the recirculation forces are greater on the upper shroud of the impeller moving it down

total thrust: sum of up and down thrust "

5. ESP components

5.1.ESP down-hole components

5.2.ESP Surface components

6. Evaluation of ESP components

6.1.ESP advantages

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General advantages of using ESP units can be summed up as follows, based on [9–11]:

- Ideally suited to produce high to extremely high liquid volumes from medium depths. Maximum rate is around 30,000 bpd from 1,000 ft.
- Energy efficiency is relatively high (around 50%) for systems producing over 1,000 bpd.
- Can be used in deviated wells without any problems.
- Requires low maintenance, provided the installation is properly designed and operated.
- Can be used in urban locations since surface equipment requires minimal space.
- Well suited to the offshore environment because of the low space requirements.
- Corrosion and scale treatments are relatively easy to perform."

6.2.ESP disadvantages

source ESP manual pdf"

General disadvantages are listed below:

- A reliable source of electric power of relatively high voltage must be available.
- The flexibility of ESP systems running on a constant electrical frequency is very low because the centrifugal pump's liquid producing capacity practically cannot be changed. Proper installation design based on accurate well inflow data and matching the unit's capacity to well deliverability is crucial, otherwise costly workover operations are required to run a new unit in the well. The use of variable speed drives can eliminate most of these problems but at an extra cost.
- Free gas present at suction conditions deteriorates the submersible pump's efficiency and can even totally prevent liquid production. The use of gas separators or gas handlers is required if more than 5% of free gas enters the pump.
- Sand or abrasive materials in well fluids increase equipment wear. Special abrasion-resistant materials are available but increase capital costs.
- Repair of ESP equipment in oilfield conditions is difficult, faulty equipment must be sent to the manufacturer's repair shop.
- High well temperature is a limiting factor, standard equipment is limited to about 250

°F, and use of special materials increases the temperature limit to 400

°F.

- Production of high viscosity oils increases power requirements and reduces lift.
- Running and pulling costs are high because of the need for heavy workover rigs. Cable suspended or CT (coiled tubing) deployed ESP units reduce workover"

7. Conclusion

Chapter4: Case study

1. Introduction

2. Hand sizing

2.1.the 9 step procedure

2.2.Hand sizing application

3. Conclusion

Chapter5: Software design

Chapter6: economic study

1.1.Why ESP ?

1.2.Technical scoring

1.3.Net Present Value (NPV) Analysis Results

Conclusion

Appendices

References

http://www.opec.org/opec_web/static_files_project/media/downloads/publications/ChildrenBook2013.pdf

<http://www.britannica.com/science/seismic-survey>

<http://ptgmedia.pearsoncmg.com/images/9780137031580/samplepages/0137031580.pdf>

http://petrowiki.org/PEH%3AInflow_and_Outflow_Performance#The_Production_System

http://www.glossary.oilfield.slb.com/Terms/b/bubble_point.aspx